

Gauge Freedom in Discrete Stokes Structures: Affine Pairings, Intrinsic Nodes, and the Rigidity of Gauss–Legendre Locality

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Abstract

We define a discrete Stokes structure by a surjection $D : E \rightarrow F$ and a symmetric boundary form b . Compatible mixed pairings exist exactly when $b|_{\ker D \times \ker D} = 0$; when nonempty, they form an affine space directed by $\Lambda^2(E/\ker D)^*$. We derive the full-rank criterion and classify left-radical lines for $\dim \ker D = 1$ and $\text{rank } b = 2$.

For the polynomial complex $\mathbb{P}_N \xrightarrow{d/dx} \mathbb{P}_{N-1}$, we determine the exact image, topology, and fibres of the projective kernel map. Interior simple roots are admissible exactly when $\sum_i \text{artanh}(x_i) = 0$. Each full-rank fibre has dimension $(N-1)(N-2)/2$ and, for $N \geq 3$, two Pfaffian components.

Scalar localization forces exactness through degree $2N-1$ and uniquely selects the zero gauge, Gauss–Legendre nodes, and positive Gauss weights. Thus Gauss–Legendre is rigid under locality but not fixed by Stokes alone.

With the L^2 quotient metric, the canonical prescribed-kernel representative has rank N in exact degree and $N-2$ otherwise. The infimum over the full-rank fibre is attained if and only if the kernel has degree N , and the lower-degree defect opens linearly under a Legendre perturbation. A global continuous, equivalently smooth, full-rank synthesis exists exactly when N is odd or $N \in \{2, 4, 8\}$; otherwise an explicit two-chart Rodrigues atlas is optimal. In even degree, every full-rank synthesis diverges near the deleted constant class. We relate the gauge to rectangular SBP and quotient-compatible skew operators on $H^1/\mathbb{R}$, while separating it from local $P^{(0)}$ terms, boundary-port changes, and weighted metrics.

Keywords. discrete Stokes structure; Stokes–Dirac structure; mixed pairing; gauge freedom; kernel polynomial; Gauss–Legendre quadrature; summation by parts; port-Hamiltonian system; orthogonal polynomials.

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1 Introduction

Stokes’ theorem fixes a boundary power balance, but it does not by itself prescribe a metric, a quadrature rule, or a nodal grid. This distinction is fundamental in open port-Hamiltonian systems. The Stokes–Dirac structure specifies the power-conserving interconnection and its boundary ports, whereas the Hamiltonian density or constitutive map specifies how energy variables are converted into efforts [26, 20].

The geometric pseudospectral construction of Moulla, Lefèvre, and Maschke [22] realizes this separation with two polynomial spaces of unequal dimensions: efforts belong to $\mathbb{P}_N$, flows to $\mathbb{P}_{N-1}$, differentiation is exact between them, and the power pairing is represented by a rectangular matrix. The construction is valid in arbitrary unisolvent flow coordinates. For the exact unweighted L^2 pairing, its one-dimensional radical is generated by the degree- N Legendre polynomial, and,

as proved below, a same-grid diagonal power chart is possible precisely at the Gauss–Legendre nodes.

Power-preserving projection and dual-representation choices also occur in weak-form and dual-field finite-element discretizations of Stokes–Dirac systems [19, 6]. Those works construct particular geometric discretizations. The question addressed here is complementary: with the length-one Stokes data (E, F, D, b) held fixed, classify the entire antisymmetric bilinear freedom and determine how it moves the polynomial radical.

The present paper asks a different question. Instead of fixing the L^2 pairing, we retain only the exact Stokes identity and classify every mixed pairing compatible with it. We first solve this problem for an arbitrary finite-dimensional surjective complex $E \xrightarrow{D} F$ with a boundary form, and only then specialize to polynomials. The difference between two compatible pairings, after composition with D , is a two-form on $E/\ker D$. We call this freedom a *pairing gauge*. The term is used in the precise linear-algebraic sense of a closed two-form or B -field transform; it should not be confused with a change of polynomial basis, with the gauge symmetry of physical potentials, or with a change of Hamiltonian density.

Once the gauge is exposed, the radical becomes gauge-dependent. Its generator κ_β may have roots different from the Legendre points. The central inverse problem is therefore:

Which projective polynomials, and which real node configurations, can occur as the radical of a full-rank Stokes-compatible mixed pairing?

We solve this problem exactly. The image lies in a projective hyperplane and is obtained by removing explicit algebraic degeneracy loci: endpoint-vanishing lines, and, in even degree, the constant line. It is therefore a relative semialgebraic open subset of that hyperplane, not the whole hyperplane. On the degree- N stratum, the corresponding root sets satisfy one nonlinear but elementary balance law. This produces a codimension-one family of admissible grids, rather than all grids.

The second question is whether these roots are quadrature nodes in the classical sense. The answer is negative in general and rigid in the only positive case. Every full-rank pairing can be represented at its distinct kernel roots by an invertible matrix of nodal couplings. That matrix is diagonal if and only if the gauge vanishes. Stokes compatibility and scalar locality together force exactness through degree $2N - 1$, and hence force the ordinary Gauss–Legendre rule. This result isolates the precise content of Gauss–Legendre canonicity: it is not absolute, but it is rigid once the unweighted Stokes geometry and local scalar power evaluation are required.

The logical dependence of the result is

$$\begin{array}{l} \boxed{\text{Stokes data}} \implies \boxed{\text{affine pairing class}} \implies \boxed{\text{intrinsic radical}} \implies \boxed{\text{full nodal matrix}}, \\ \boxed{\text{scalar locality}} \implies \boxed{L^2} \implies \boxed{\text{Gauss–Legendre}}. \end{array} \tag{1}$$

The first chain holds throughout the full-rank stratum; the second is the additional rigidity statement.

The main contributions are as follows.

1. We prove the abstract affine-classification theorem, a necessary and sufficient full-rank existence criterion, and a radical-line theorem for rank-two boundary forms.
2. We determine the exact polynomial kernel image, its topology, construct every admissible line, and compute both the dimension and topology of every full-rank fibre. We also classify the orders admitting a global continuous or smooth full-rank synthesis over the whole image, determine the minimal atlas size, give an explicit two-chart formula, and prove the unavoidable radial blow-up near the deleted constant class.
3. We characterize the simple interior root sets by a hyperbolic barycentre condition and separate intrinsic kernel nodes, full matrix localization, and scalar quadrature.

4. We formulate positive localization as a truncated moment problem, use the differentiated product span to prove Gauss–Legendre rigidity, and record the independent Tchakaloff existence mechanism for positive moment functionals.
5. Relative to the L^2 quotient metric, we derive the closed formula for the minimum-norm solution of a prescribed-kernel constraint and prove its global rank stratification: its rank is N for degree- N kernels and exactly $N - 2$ on every nonempty lower-degree stratum, where both rectangular radicals are computed explicitly. We prove that the full-rank minimum is attained exactly in the degree- N stratum, that the lower defect opens transversely in the Legendre direction with two-sided singular-value bounds, and then establish local stability near Gauss and show why no uniform global estimate can ignore node collision.
6. We give the precise rectangularization statement for SBP, an exact continuous operator model with a coherent-extension criterion, and a careful comparison with Dirac two-form gauges, $P^{(0)}$ freedom, boundary-port coordinates, and weighted metrics.

The relation between SBP and quadrature is classical [15, 10]; function-space SBP extends the theory to non-polynomial spaces [12], and recent work constructs minimal FSBP operators through generalized Gaussian or Gauss–Lobatto rules [14, 3]. Gauge equivalence of Dirac structures by closed two-forms is also established in Dirac geometry [8]. The contribution claimed here is narrower and different: an exact classification of the abstract affine pairing freedom and, in the polynomial instance, of the projective kernel map, together with the locality rigidity and local stability theorems.

2 Abstract discrete Stokes structures

2.1 Definition and affine classification

Let E and F be finite-dimensional real vector spaces, let $D : E \rightarrow F$ be surjective, and let $b \in S^2 E^*$ be a symmetric bilinear form. Put $K = \ker D$ and $m = \dim F$.

Definition 2.1 (Discrete Stokes structure and compatible pairing). The quadruple (E, F, D, b) is called a *discrete Stokes structure*. A mixed bilinear pairing $M : E \times F \rightarrow \mathbb{R}$ is *Stokes-compatible* if

$$M(e, De') + M(e', De) = b(e, e'), \quad e, e' \in E. \quad (2)$$

The set of all such pairings is denoted by $\mathcal{C}(D, b)$. A compatible pairing is *full rank* when the map $M^\flat : E \rightarrow F^*$, $e \mapsto M(e, \cdot)$, has rank $m = \dim F$.

Associate with M the bilinear form

$$a_M(e, e') = M(e, De'). \quad (3)$$

It satisfies $a_M + a_M^\top = b$ and $a_M(E, K) = 0$.

Theorem 2.2 (Abstract affine classification). *The class $\mathcal{C}(D, b)$ is nonempty if and only if*

$$b|_{K \times K} = 0. \quad (4)$$

When this condition holds, $\mathcal{C}(D, b)$ is an affine space modelled on

$$\Lambda^2(E/K)^* \simeq \Lambda^2 F^*, \quad (5)$$

and therefore

$$\dim \mathcal{C}(D, b) = \frac{m(m-1)}{2}. \quad (6)$$

Proof. If M is compatible and $k, k' \in K$, then both terms on the left of (2) vanish, hence $b(k, k') = 0$.

Conversely, choose a splitting $E = K \oplus H$. Define $a : E \times E \rightarrow \mathbb{R}$ by

$$a(e, k) = 0, \quad a(k, h) = b(k, h), \quad a(h, h') = \frac{1}{2}b(h, h')$$

for $e \in E$, $k \in K$, and $h, h' \in H$. The assumption on b gives $a + a^\top = b$, and $a(E, K) = 0$. Therefore

$$M(e, f) = a(e, h_f),$$

where $h_f \in H$ is the unique vector satisfying $Dh_f = f$, defines a compatible pairing.

If M_1, M_0 are compatible, then $\omega = a_{M_1} - a_{M_0}$ is skew-symmetric and $\omega(E, K) = 0$. Skew-symmetry also gives $\omega(K, E) = 0$, so ω descends uniquely to $\Lambda^2(E/K)^*$. Conversely, every such two-form may be added to a_{M_0} . Finally $E/K \simeq F$ because D is surjective, which proves (6). $\square$

The theorem separates an existence obstruction from the gauge freedom. It also specifies which radical statement is valid: K is always contained in the right radical of a_M , but

$$a_M(k, e) = b(k, e), \quad k \in K, e \in E, \quad (7)$$

so K need not lie in the left radical.

Theorem 2.3 (Existence of full-rank compatible pairings). *Assume $b|_{K \times K} = 0$, so that $\mathcal{C}(D, b)$ is nonempty. A full-rank compatible pairing exists if and only if*

$$\boxed{m \text{ is even} \quad \text{or} \quad b \neq 0.} \quad (8)$$

Equivalently, the only obstruction beyond compatibility is the boundary-free odd-dimensional case $b = 0$ and m odd.

Proof. Choose a splitting $E = K \oplus H$ and identify $H \simeq F$. Relative to this splitting, every compatible form a_M has a matrix

$$[a_M] = \begin{pmatrix} 0 & B \\ 0 & S + \Omega \end{pmatrix}, \quad S^\top = S, \quad \Omega^\top = -\Omega, \quad (9)$$

where $[b] = \begin{pmatrix} 0 & B \\ B^\top & 2S \end{pmatrix}$. The ranks of M and a_M agree because D is onto. Hence M is full rank precisely when

$$H \longrightarrow K^* \oplus H^*, \quad h \longmapsto (Bh, (S + \Omega)h) \quad (10)$$

is injective.

If m is even, choose a nondegenerate skew form Ω_0 on H . Then $S + t\Omega_0$ is invertible for all sufficiently large $|t|$, so (10) is injective. Let now m be odd and $b \neq 0$. If $B \neq 0$, choose $v \in H$ with $Bv \neq 0$ and a skew form Ω_0 with $\ker \Omega_0 = \text{span}\{v\}$. The map obtained from $\Omega = t\Omega_0$ is injective for all sufficiently large $|t|$: otherwise unit null vectors along a sequence $|t| \rightarrow \infty$ would converge to a nonzero multiple of v while still belonging to $\ker B$, a contradiction. If $B = 0$, then $S \neq 0$. Choose v with $S(v, v) \neq 0$ and again $\ker \Omega_0 = \text{span}\{v\}$. A hypothetical sequence of unit vectors in $\ker(S + t\Omega_0)$ converges to a multiple of v ; pairing the kernel equation with v and using $\Omega_0 v = 0$ contradicts $S(v, v) \neq 0$.

Finally, if $b = 0$, then $B = S = 0$ and (10) is injective exactly when Ω is nondegenerate. No skew form is nondegenerate in odd dimension. This proves both necessity and sufficiency. $\square$

Corollary 2.4 (Genericity of full rank). *Assume $b|_{K \times K} = 0$ and that the criterion in Theorem 2.3 holds. Then the full-rank pairings form a nonempty relative Zariski-open subset of the affine space $\mathcal{C}(D, b)$; in particular, they are Euclidean dense.*

Proof. In affine coordinates on $\mathcal{C}(D, b)$, failure of full rank is the common zero set of the maximal minors of the matrix of M . Its complement is therefore relative Zariski open. It is nonempty by Theorem 2.3; since an affine space is irreducible, every nonempty relative Zariski-open subset is Zariski dense, hence Euclidean dense. $\square$

Corollary 2.5 (Boundary-free parity obstruction). *Assume $b = 0$. Then a_M descends to a skew form $\bar{a}_M$ on E/K . The mixed pairing has rank m if and only if $\bar{a}_M$ is nondegenerate. In particular, a full-rank compatible pairing exists if and only if m is even.*

Proof. For $b = 0$, equation (7) puts K in both radicals and a_M descends to E/K . Since $D^* : F^* \rightarrow E^*$ is injective, M and a_M have the same rank. Thus maximal rank is equivalent to nondegeneracy of the descended skew form. Such a form exists exactly in even dimension. $\square$

The next result isolates the mechanism that will later produce the polynomial endpoint condition. For a bilinear form a , write $\text{rad}_L a = \{e \in E : a(e, E) = 0\}$.

Theorem 2.6 (Radical lines for a rank-two boundary). *Assume $b|_{K \times K} = 0$, $\dim K = 1$, $\text{rank } b = 2$, and $b(K, E) \neq 0$. Fix $e \in K \setminus \{0\}$ and put $Z = \text{rad } b$. Then the left radical of every full-rank compatible pairing is a line. Moreover, the set of all such lines is exactly*

$$m \text{ even: } \{L \in \mathbb{P}(E) : L \subset e^{\perp_b}, L \neq K, L \not\subset Z\}, \quad (11)$$

$$m \text{ odd: } \{L = \text{span}\{\kappa\} : b(\kappa, \kappa) = 0, b(e, \kappa) \neq 0\}. \quad (12)$$

Every line displayed in (11)–(12) is realized constructively by a full-rank compatible pairing.

Proof. Rescale a vector h so that $b(e, h) = 2$, and replace h by a multiple of e plus h so that $b(h, h) = 0$. Then $E = \text{span}\{e, h\} \oplus Z$, $\dim Z = m - 1$, and every compatible form has the unique matrix

$$A_{r,C} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & r^\top \\ 0 & -r & C \end{pmatrix}, \quad C^\top = -C. \quad (13)$$

It has rank m exactly when $\ker C \cap r^\perp = \{0\}$. If $\kappa = \alpha e + \beta h + z$, its left-kernel equations are

$$z^\top r = 2\alpha, \quad Cz = \beta r. \quad (14)$$

If m is even, then $\dim Z$ is odd. Full rank forces $\ker C = \text{span}\{z\}$ and $z^\top r \neq 0$; equation (14) gives $\beta = 0$, $z \neq 0$, and $\alpha \neq 0$. This is exactly (11). Conversely, for $\kappa = \alpha e + z$ satisfying those conditions, choose a skew C with kernel $\text{span}\{z\}$ and choose r with $z^\top r = 2\alpha$.

If m is odd, then $\dim Z$ is even and full rank forces C to be invertible. Equations (14) give $\beta \neq 0$ and $\alpha = \frac{1}{2}z^\top r = \frac{1}{2\beta}z^\top Cz = 0$. Since $b(\kappa, \kappa) = 4\alpha\beta$ and $b(e, \kappa) = 2\beta$, this is precisely (12). Conversely, write any such vector as $\kappa = \beta h + z$, choose an invertible skew C on Z , and set $r = \beta^{-1}Cz$. In both parities the constructed matrix has rank m and the prescribed left radical. $\square$

Remark 2.7 (Examples and limits of the abstraction). The polynomial construction below has $K = \mathbb{R}$ and an endpoint boundary form. A finite spline space with $F = DE$ gives the same one-dimensional kernel and boundary trace. For periodic trigonometric polynomials, $F = DE$ is the zero-mean space, $b = 0$, and $\dim F$ is even, so Corollary 2.5 does not forbid full rank. For graph cochains, the incidence map is generally not onto the full edge space when cycles are present; the theorem applies directly to $F = \text{im } D$, while a treatment retaining the cycle space requires the longer complex and its cohomology. These examples show both the reach and the precise scope of the length-one theory.

2.2 Scalar localization requires additional moment data

The quadruple (E, F, D, b) contains no points, evaluation maps, or multiplication. Locality therefore cannot be defined from the Stokes structure alone. Suppose additionally that E and F are function spaces on a set X , and put

$$G = \text{span}\{ef : e \in E, f \in F\}.$$

For a linear functional $L \in G^*$, define the product pairing

$$M_L(e, f) = L(ef). \quad (15)$$

Let

$$\delta : S^2 E \longrightarrow G, \quad \delta(e, e') = eDe' + e'De. \quad (16)$$

Proposition 2.8 (Moment-factorization criterion). *Assume that a reference functional $L_0 \in G^*$ satisfies $L_0(\delta(e, e')) = b(e, e')$. Then the Stokes-compatible product pairings are precisely*

$$L_0 + (\text{im } \delta)^\perp \subset G^*. \quad (17)$$

In particular, the product pairing is unique if and only if $\text{im } \delta = G$. An atomic scalar localization with r nodes and positive weights exists precisely when the corresponding L has a representation

$$L = \sum_{i=1}^r w_i \text{ev}_{x_i}|_G, \quad x_i \in X, \quad w_i > 0, \quad (18)$$

which is a truncated moment problem on G .

Proof. Substitution of (15) into (2) gives $L(\delta(e, e')) = b(e, e')$. Subtracting the same identity for L_0 proves (17); the remaining statements are immediate. $\square$

Proposition 2.9 (Positive localization by Tchakaloff). *Let X be compact, let $G \subset C(X)$ be finite dimensional and contain the constants, and let $L \in G^*$ be representable by a finite positive Borel measure on X . Then there exist $r \leq \dim G$, nodes $x_i \in X$, and positive weights w_i such that*

$$L(g) = \sum_{i=1}^r w_i g(x_i), \quad g \in G. \quad (19)$$

Consequently, whenever the compatible moment slice is nonempty, its positive localizable part is

$$(L_0 + (\text{im } \delta)^\perp) \cap \mathcal{M}_+(G), \quad (20)$$

where $\mathcal{M}_+(G)$ denotes the cone of functionals represented by positive measures; every element of this intersection has a positive atomic representative with at most $\dim G$ nodes.

Proof. Choose a basis $g_1, \dots, g_s$ of G with $g_1 = 1$ and consider the continuous moment map $\Gamma(x) = (g_1(x), \dots, g_s(x))$. The moment vector of L lies in the conic hull of $\Gamma(X)$. The conic Carathéodory theorem represents it as a positive combination of at most $s = \dim G$ such vectors, giving (19). Formula (20) then follows from Proposition 2.8. This is Tchakaloff's theorem in the present finite-dimensional form [25, 2]. $\square$

Complete Chebyshev systems provide classical hypotheses under which generalized Gaussian representations exist and are unique; weak Chebyshev systems lead to related moment and spline quadrature theories [17, 13, 21, 9]. These results concern the additional moment cone in (18), not the bare Stokes quadruple. Consequently, a classification of all function spaces admitting a minimal positive localization is a separate moment-theoretic problem rather than a corollary of Theorem 2.2.

Thus three levels must be kept separate: affine compatibility of a mixed pairing, positive representability of a moment functional, and existence or uniqueness of a finite atomic representation. Tchakaloff shows that every positively representable functional admits a positive atomic representation with at most $\dim G$ atoms. It neither proves that the compatible affine slice meets the positive moment cone nor guarantees a minimal node count; Chebyshev-system theory supplies sharper counts and uniqueness only under additional structure.

3 Polynomial mixed Stokes pairings

3.1 Spaces, derivative, and boundary form

Fix $N \geq 1$ and set

$$E_N = \mathbb{P}_N[-1, 1], \quad F_N = \mathbb{P}_{N-1}[-1, 1].$$

Throughout the polynomial sections, P_n denotes the Legendre polynomial normalized by $P_n(1) = 1$. The derivative

$$d : E_N \longrightarrow F_N, \quad dq = q',$$

is surjective and has kernel $\mathbb{R}$. Define the symmetric boundary form

$$b(p, q) = [pq]_{-1}^1 = p(1)q(1) - p(-1)q(-1). \quad (21)$$

Definition 3.1 (Stokes-compatible mixed pairing). A bilinear map $\beta : E_N \times F_N \rightarrow \mathbb{R}$ is *Stokes-compatible* if

$$\beta(p, q') + \beta(q, p') = b(p, q), \quad p, q \in E_N. \quad (22)$$

It is *full rank* if the map

$$\beta^\flat : E_N \rightarrow F_N^*, \quad p \mapsto \beta(p, \cdot),$$

has rank N .

In bases of E_N and F_N , let $M \in \mathbb{R}^{(N+1) \times N}$ represent β and $D \in \mathbb{R}^{N \times (N+1)}$ represent d . If R evaluates at the endpoints and $\Sigma = \text{diag}(-1, 1)$, then (22) is exactly

$$MD + D^\top M^\top = R^\top \Sigma R. \quad (23)$$

This is the rectangular discrete Stokes identity of the 2012 construction [22].

The reference pairing is

$$\beta_0(p, r) = \int_{-1}^1 p(x)r(x) \, dx. \quad (24)$$

Integration by parts shows that β_0 is Stokes-compatible.

3.2 From a rectangular pairing to a bilinear form on efforts

Associate with β the form

$$a_\beta(p, q) = \beta(p, q'), \quad p, q \in E_N. \quad (25)$$

It satisfies

$$a_\beta(p, q) + a_\beta(q, p) = b(p, q), \quad a_\beta(p, 1) = 0. \quad (26)$$

Conversely, every bilinear form satisfying (26) determines a unique mixed pairing. Indeed, for $r \in F_N$, choose any primitive $q \in E_N$ with $q' = r$ and set $\beta(p, r) = a_\beta(p, q)$. This is independent of the additive constant in q .

For the reference pairing,

$$a_0(p, q) = \int_{-1}^1 p(x)q'(x) \, dx.$$

The symmetric part of a_0 is the boundary form, not the L^2 product. The role of L^2 is that of a distinguished reference point in the affine class.

4 Affine classification and pairing gauge

Theorem 4.1 (Affine classification). *The set $\mathcal{A}_N$ of Stokes-compatible mixed pairings is an affine space with origin β_0 and model vector space*

$$\Lambda^2(E_N/\mathbb{R})^*.$$

Equivalently, every compatible form has a unique decomposition

$$a_\beta = a_0 + \omega, \quad \omega \in \Lambda^2(E_N/\mathbb{R})^*. \quad (27)$$

Consequently,

$$\dim \mathcal{A}_N = \frac{N(N-1)}{2}. \quad (28)$$

Proof. Let β_1 and β_0 be compatible and put $\omega = a_{\beta_1} - a_{\beta_0}$. Subtracting their Stokes identities gives

$$\omega(p, q) + \omega(q, p) = 0.$$

Thus ω is skew-symmetric. Since both forms vanish in the second slot on constants, $\omega(p, 1) = 0$; skew-symmetry also gives $\omega(1, p) = 0$. Hence ω descends to a two-form on $E_N/\mathbb{R}$.

Conversely, if $\omega \in \Lambda^2(E_N/\mathbb{R})^*$, then $a_0 + \omega$ vanishes on constants in the second slot and has symmetric part b . It therefore defines a unique compatible mixed pairing. Since $\dim(E_N/\mathbb{R}) = N$, the dimension is $N(N-1)/2$. $\square$

Definition 4.2 (Pairing gauge). The two-form ω in (27) is called the *pairing gauge*. The reference $\omega = 0$ is the L^2 gauge.

The terminology records a power-invisible antisymmetric freedom. It is not a basis change: different ω define different bilinear pairings. Since $E_N/\mathbb{R}$ is a vector space and ω is constant, it is also a closed two-form, which anticipates the Dirac-geometric interpretation in Section 13.

In matrix form, if M_0 represents β_0 and $\Delta M = M - M_0$, then

$$\Omega = \Delta M D \quad (29)$$

represents ω . Equation (23) gives $\Omega^\top = -\Omega$, and $D1 = 0$ gives $\Omega 1 = 0$.

5 Canonical normal form and the full-rank locus

Assume $N \geq 2$. Introduce

$$e = 1, \quad h = x, \quad Z_N = (1 - x^2)\mathbb{P}_{N-2}.$$

Then

$$E_N = \text{span}\{e, h\} \oplus Z_N, \quad d_Z := \dim Z_N = N - 1. \quad (30)$$

Every element is written uniquely as $p = \alpha e + \beta h + z$, with $z \in Z_N$. In this decomposition, the boundary form has matrix

$$[b] = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (31)$$

Proposition 5.1 (Canonical normal form). *Every Stokes-compatible form a_β has a unique matrix*

$$A_{r,C} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & r^\top \\ 0 & -r & C \end{pmatrix}, \quad r \in \mathbb{R}^{dz}, \quad C \in \mathbb{R}^{dz \times dz}, \quad C^\top = -C. \quad (32)$$

Conversely, every matrix (32) defines a Stokes-compatible pairing. Moreover,

$$\text{rank } A_{r,C} = N \iff \ker C \cap r^\perp = \{0\}. \quad (33)$$

Proof. The first column of A is zero because $a_\beta(\cdot, 1) = 0$. The equation $A + A^\top = [b]$ then fixes the first row, the 2×2 block, and the off-diagonal signs, leaving precisely a vector r and a skew matrix C .

For $u = u_e e + u_h h + u_z$, the equation $A_{r,C} u = 0$ gives

$$u_h = 0, \quad C u_z = 0, \quad r^\top u_z = 0.$$

Thus the right kernel is $\text{span}\{e\} \oplus (\ker C \cap r^\perp)$. Since the maximal possible rank is N , the rank criterion follows. $\square$

Because D is surjective, $\text{rank}(MD) = \text{rank } M$. Hence full rank of β is equivalent to $\text{rank } A_{r,C} = N$. For a full-rank pairing the left radical is one-dimensional. If

$$\kappa = \alpha e + \beta h + z, \quad z \in Z_N, \quad (34)$$

then $\kappa^\top A_{r,C} = 0$ is equivalent to

$$z^\top r = 2\alpha, \quad C z = \beta r. \quad (35)$$

6 The kernel map and its exact image

Let $\mathcal{A}_N^{\text{fr}} \subset \mathcal{A}_N$ denote the open subset of full-rank pairings. Define the projective kernel map

$$\mathcal{K}_N : \mathcal{A}_N^{\text{fr}} \longrightarrow \mathbb{P}(E_N), \quad \beta \longmapsto [\ker \beta^b] = [\kappa_\beta]. \quad (36)$$

Theorem 6.1 (Exact image of the kernel map). *For $N \geq 1$, the image of $\mathcal{K}_N$ consists exactly of the projective classes of nonconstant polynomials $\kappa \in E_N$ satisfying*

$$\boxed{\kappa(1) = (-1)^N \kappa(-1) \neq 0.} \quad (37)$$

In particular, the degree- N stratum of the image is precisely the set of degree- N polynomials satisfying (37). If

$$\mathcal{H}_N = \mathbb{P}\left(\ker[\kappa \mapsto \kappa(1) - (-1)^N \kappa(-1)]\right), \quad (38)$$

then $\text{im } \mathcal{K}_N$ is the relative semialgebraic open subset of $\mathcal{H}_N$ obtained by removing the endpoint-vanishing locus and, when N is even, the constant line.

Proof. Here $K = \mathbb{R}e$, $m = N$, $\text{rank } b = 2$, and $b(e, p) = p(1) - p(-1)$. If N is even, Theorem 2.6 says that the radical line lies in $e^\perp$, is not the constant line, and is not contained in $\text{rad } b = Z_N$. These three conditions are respectively

$$\kappa(1) = \kappa(-1), \quad \kappa \text{ nonconstant}, \quad \kappa(1) = \kappa(-1) \neq 0.$$

If N is odd, the abstract conditions become

$$\kappa(1)^2 - \kappa(-1)^2 = 0, \quad \kappa(1) - \kappa(-1) \neq 0,$$

which are equivalent to $\kappa(1) = -\kappa(-1) \neq 0$. The constructive converse in Theorem 2.6 proves sufficiency in both parities. Finally, the excluded conditions are algebraic closed subsets of the projective hyperplane (38); their complement is therefore relative semialgebraic open. $\square$

Remark 6.2 (Degree drop). Full rank does not force $\deg \kappa_\beta = N$. Lower-degree kernel polynomials occur on proper algebraic strata of the image. The node theorem below therefore concerns the open degree- N stratum. This distinction is essential: only a degree- N kernel can supply N finite roots counted with multiplicity.

The endpoint value in Theorem 6.1 gives every projective class in the image a unique representative satisfying $\kappa(1) = 1$. This normalization reveals the topology hidden by the projective notation.

Theorem 6.3 (Topology of the kernel image). *Let $\mathcal{B}_N = \text{im } \mathcal{K}_N$, represented by the normalized polynomials $\kappa(1) = 1$. Then*

$$\mathcal{B}_N \simeq \begin{cases} \mathbb{R}^{N-1}, & N \text{ odd}, \\ \mathbb{R}^{N-1} \setminus \{0\}, & N \text{ even}. \end{cases} \quad (39)$$

More precisely, for $N \geq 2$, with $Z_N = (1 - x^2)\mathbb{P}_{N-2}$,

$$\kappa = x + z \quad (N \text{ odd}), \quad \kappa = 1 + z, \quad z \neq 0 \quad (N \text{ even}). \quad (40)$$

Thus $\mathcal{B}_N$ is contractible for odd N ; for $N = 2$ it has two components; and for even $N \geq 4$ it is connected and deformation retracts onto S^{N-2} .

For $N \geq 2$, the exact-degree stratum $\mathcal{B}_N^{(N)} = \{[\kappa] \in \mathcal{B}_N : \deg \kappa = N\}$ has exactly two contractible components, distinguished projectively by the sign of

$$c_N(\kappa)\kappa(1), \quad (41)$$

where $c_N(\kappa)$ is the coefficient of P_N in the Legendre expansion of κ . The chamber of polynomials with N simple roots in $(-1, 1)$ lies in the positive component. For $N = 1$, the base and its exact-degree stratum are the single class $[x]$.

Proof. For $N = 1$ the base is $[x]$. Assume $N \geq 2$. In the decomposition $E_N = \text{span}\{1, x\} \oplus Z_N$, the normalized endpoint condition forces respectively the two representations in (40). Since $\dim Z_N = N - 1$, this proves (39) and its stated homotopy consequences.

For $N \geq 2$, exact degree is the complement of the affine hyperplane $c_N(\kappa) = 0$ in these coordinates. Its two open half-spaces are contractible. Under a projective rescaling both factors in (41) acquire the same scalar, so its sign is intrinsic. If κ has simple interior roots, its monic representative has positive leading Legendre coefficient and positive value at 1; hence the sign is positive. This completes the proof. $\square$

6.1 Real node configurations

Let κ be monic of degree N with roots $x_1, \dots, x_N$. Then

$$\kappa(1) = \prod_{i=1}^N (1 - x_i), \quad (-1)^N \kappa(-1) = \prod_{i=1}^N (1 + x_i).$$

Theorem 6.1 immediately gives the following classification.

Corollary 6.4 (Admissible interior nodes). *Let $-1 < x_1 < \dots < x_N < 1$. There exists a full-rank Stokes-compatible pairing whose kernel polynomial has precisely these roots if and only if*

$$\prod_{i=1}^N (1 - x_i) = \prod_{i=1}^N (1 + x_i). \quad (42)$$

Equivalently,

$$\sum_{i=1}^N \text{artanh}(x_i) = 0. \quad (43)$$

Proof. The product identity is the endpoint condition. Since all factors are positive, taking logarithms and using $2 \operatorname{artanh} x = \log((1+x)/(1-x))$ gives (43). Sufficiency follows from Theorem 6.1. $\square$

Let

$$\mathfrak{C}_N = \{(x_1, \dots, x_N) : -1 < x_1 < \dots < x_N < 1\}.$$

The map $\Phi(x) = \sum_i \operatorname{artanh}(x_i)$ has nonzero gradient

$$\nabla \Phi(x) = \left(\frac{1}{1-x_1^2}, \dots, \frac{1}{1-x_N^2} \right).$$

Hence the admissible ordered node configurations form the smooth hypersurface $\Phi^{-1}(0) \subset \mathfrak{C}_N$, of dimension $N-1$. Stokes therefore imposes a genuine codimension-one obstruction: it does not generate arbitrary grids.

Proposition 6.5 (Hyperbolic barycentre and configuration geometry). *Equip $I = (-1, 1)$ with the one-dimensional Klein–Hilbert metric*

$$g_I = \frac{dx^2}{(1-x^2)^2}. \quad (44)$$

Then $y = \operatorname{artanh} x$ is a global geodesic coordinate and an isometry from (I, g_I) to the Euclidean line. For $X = (x_1, \dots, x_N) \in \mathfrak{C}_N$, condition (43) is equivalent to saying that the equal-weight Fréchet barycentre of $x_1, \dots, x_N$ is the origin. In product rapidity coordinates, the admissible configuration space is

$$\left\{ y_1 < \dots < y_N : \sum_{i=1}^N y_i = 0 \right\}, \quad (45)$$

an open chamber in a totally geodesic hyperplane of $\mathbb{R}^N$.

Proof. Equation (44) gives $ds = dx/(1-x^2) = dy$. Hence the Fréchet functional of a candidate barycentre with rapidity z is $\sum_i (z - y_i)^2$, whose unique minimizer is $z = N^{-1} \sum_i y_i$. It is the origin exactly under (43). The product metric becomes Euclidean in the y_i , proving the last statement. $\square$

Remark 6.6 (What the hyperbolic interpretation does not assert). The endpoints ± 1 are the two ideal points of the interval model, and the Stokes form uses traces at those endpoints. This observation does not by itself make the polynomial Stokes structure equivariant under the hyperbolic isometry group: polynomial spaces are not preserved by general Möbius reparametrizations. Nor does Proposition 6.5 produce a multivariate Gaussian cubature on the Poincaré disk. Such an extension would require a separate differential complex, boundary trace, moment space, and zero-set theory.

6.2 Fibres of the kernel map

Theorem 6.7 (Fibre dimension). *For every projective kernel class in the image of $\mathcal{K}_N$, the corresponding full-rank fibre is a nonempty smooth open subset of an affine space of dimension*

$$\boxed{\frac{(N-1)(N-2)}{2}}. \quad (46)$$

Proof. Use the decomposition $\kappa = \alpha e + \beta h + z$.

For even N , $\beta = 0$, $z \neq 0$. The condition $Cz = 0$ leaves a skew form on the $(d_Z - 1)$ -dimensional quotient $Z_N / \operatorname{span}\{z\}$, of dimension $(d_Z - 1)(d_Z - 2)/2$. The condition $z^\top r = 2\alpha$ leaves $d_Z - 1$

free components of r . The total is $d_Z(d_Z - 1)/2 = (N - 1)(N - 2)/2$. Full rank is the open condition that the induced skew form on the even-dimensional quotient be nondegenerate.

For odd N , C may be any invertible skew matrix and $r = \beta^{-1}Cz$ is then fixed. Invertible skew matrices form an open subset of dimension $d_Z(d_Z - 1)/2$, giving the same formula. $\square$

The dimension identity

$$\frac{N(N - 1)}{2} = (N - 1) + \frac{(N - 1)(N - 2)}{2}$$

separates the $N - 1$ observable kernel directions from the gauge directions invisible to the kernel map.

Theorem 6.8 (Topology of the full-rank fibres). *For $N = 1$ and $N = 2$, every nonempty full-rank fibre is a point. For every $N \geq 3$, each full-rank fibre has exactly two connected components. After orienting the relevant even dimensional quotient, the components are distinguished by the sign of a Pfaffian.*

Proof. For odd $N \geq 3$, the proof of Theorem 6.7 identifies the fibre with the space of invertible skew forms C on the even-dimensional space Z_N ; r is then fixed by C . The sign of $\text{Pf}(C)$ is constant on connected components, and the two sign sets are connected because each is a homogeneous space of one component of $\text{GL}(Z_N)$ by the symplectic group.

For even N , the prescribed kernel fixes $z \neq 0$. The skew form C descends to a nondegenerate skew form on the even-dimensional quotient $Z_N/\text{span}\{z\}$, while r ranges over the affine hyperplane $z^\top r = 2\alpha$. The latter is contractible, so the two components are again the two Pfaffian signs of the quotient form. When $N = 2$ that quotient is zero dimensional and the affine condition fixes r , leaving a point. For $N = 1$ the gauge space itself is zero dimensional. $\square$

Pointwise synthesis in Theorem 6.1 does not automatically give a continuous choice of pairing over the whole base. The topology of Theorem 6.3 produces an exact obstruction.

Theorem 6.9 (Global continuous and smooth full-rank sections). *The projective kernel map*

$$\mathcal{K}_N : \mathcal{A}_N^{\text{fr}} \longrightarrow \mathcal{B}_N \quad (47)$$

admits a global continuous section if and only if it admits a global smooth section, and these equivalent conditions hold if and only if

$$\boxed{N \text{ is odd} \quad \text{or} \quad N \in \{2, 4, 8\}.} \quad (48)$$

For even N , the obstruction is equivalent to the existence of an almost complex structure on S^{N-2} .

Proof. For $N = 1$ the assertion is immediate. Suppose first that $N \geq 3$ is odd. Then $d_Z = N - 1$ is even. Fix a nondegenerate skew matrix C_0 on Z_N . For the normalized kernel $\kappa = x + z$, set

$$C_z = C_0, \quad r_z = C_0 z. \quad (49)$$

The equations (35) hold, C_0 is invertible, and Proposition 5.1 gives a full-rank pairing depending smoothly on z .

Let now N be even. Write the normalized kernel as $\kappa = 1 + z$ with $z \in Z_N \setminus \{0\}$. Equations (35) and the full-rank condition are equivalent to

$$C_z^\top = -C_z, \quad \ker C_z = \text{span}\{z\}, \quad z^\top r_z = 2. \quad (50)$$

Fix an auxiliary Euclidean inner product on Z_N , use its musical isomorphism $z \mapsto z^b$, and set

$$r_z = \frac{2z^b}{\|z\|^2}. \quad (51)$$

Thus all topology is carried by C_z . Restricting a family C_z to the unit sphere in Z_N gives the tangent two-form

$$\omega_z(v, w) = \langle v, C_z w \rangle, \quad v, w \in T_z S(Z_N) = z^\perp. \quad (52)$$

It is nondegenerate precisely because $\ker C_z = \text{span}\{z\}$. Hence a continuous section produces a continuous almost symplectic structure on

$$T_z S(Z_N) = z^\perp. \quad (53)$$

Conversely, an almost symplectic structure, together with the auxiliary metric, produces the required skew endomorphism C_z , extended by zero on $\text{span}\{z\}$ and radially on $Z_N \setminus \{0\}$. To justify smoothing, use the auxiliary metric to realize $\Lambda^2 T^* S^{N-2}$ as a smooth vector subbundle of the trivial bundle $S^{N-2} \times \Lambda^2 Z_N^*$. Approximate a continuous section as an ambient map by the relative smooth approximation theorem [16, Theorem 2.2.5], then apply the smooth fibrewise orthogonal projection back to $\Lambda^2 T^* S^{N-2}$. On the compact sphere, a continuous nondegenerate form has a positive uniform lower bound on its least tangent singular value, so every sufficiently close smooth approximation remains nondegenerate. Thus continuous and smooth existence are equivalent.

An almost symplectic structure is equivalent to an almost complex structure. For $N = 2$ the tangent bundle of S^0 is zero dimensional. Among positive-dimensional spheres, only S^2 and S^6 admit almost complex structures [4, 18]. Hence the even orders are exactly $N = 2, 4, 8$. $\square$

The obstruction concerns the whole projective kernel image, including its lower-degree strata. It disappears on the exact-degree set $\mathcal{B}_N^{(N)}$: its connected components are contractible, and Theorem 9.3 below provides a canonical full-rank section. Thus the obstruction does not restrict the admissible simple node configurations.

Proposition 6.10 (Explicit exceptional sections). *For $N = 2, 4, 8$, the even-order sections in Theorem 6.9 may be chosen explicitly. With the auxiliary metric fixed as above, write $\hat{z} = z/\|z\|$ for $z \neq 0$:*

1. for $N = 2$, take $C_z = 0$;
2. for $N = 4$, identify the oriented space Z_4 with $\mathbb{R}^3$ and set $C_z v = \hat{z} \times v$;
3. for $N = 8$, identify Z_8 with the imaginary octonions and use the seven-dimensional vector cross product $C_z v = \hat{z} \times v$.

In every case take $r_z = 2z^b/\|z\|^2$.

Proof. In dimensions three and seven, the indicated cross products satisfy $\langle \hat{z} \times v, w \rangle = -\langle v, \hat{z} \times w \rangle$ and $\ker(v \mapsto \hat{z} \times v) = \text{span}\{z\}$ for $z \neq 0$; see [5, 1]. Formula (51) supplies the remaining normal-form condition. Moreover, $C_z^2 = -(I - \hat{z} \otimes \hat{z}^b)$, so the nonzero tangent singular values equal one. The case $N = 2$ has $\dim Z_2 = 1$ and zero-dimensional tangent spaces. $\square$

We call a *synthesis chart* an open subset of $\mathcal{B}_N$ endowed with a smooth full-rank local section of $\mathcal{K}_N$.

Corollary 6.11 (Minimal synthesis atlas). *The minimum number of smooth full-rank synthesis charts covering $\mathcal{B}_N$ is one when N is odd or $N \in \{2, 4, 8\}$, and two for every other even N .*

Proof. One chart exists exactly in the orders of Theorem 6.9. In the remaining even orders, one chart is impossible. Two suffice because the sphere S^{N-2} is covered by two punctured spheres; each is contractible, so its tangent bundle admits a smooth symplectic frame. Extend the corresponding nondegenerate tangent form radially and use (51). $\square$

Remark 6.12 (Reproducible auxiliary choices). The atlas becomes deterministic once auxiliary conventions are fixed. One convenient choice is the L^2 metric on Z_N , followed by Gram–Schmidt orthonormalization of the ordered basis

$$(1 - x^2), x(1 - x^2), \dots, x^{N-2}(1 - x^2),$$

with positive normalization factors. Write the resulting oriented basis as $(e_1, \dots, e_{N-1})$, take $e = e_1$, set $J_0 e_1 = 0$, and define

$$J_0 e_{2j} = e_{2j+1}, \quad J_0 e_{2j+1} = -e_{2j}, \quad 1 \leq j \leq \frac{N-2}{2}.$$

These conventions are computational rather than intrinsic; changing them changes the local section but not the atlas size, the kernel line, or the Stokes identity.

Proposition 6.13 (Explicit two-chart synthesis). *Let N be even. Fix a unit vector $e \in Z_N$ and a skew endomorphism J_0 such that $\ker J_0 = \text{span}\{e\}$ and $J_0^2 = -I$ on $e^\perp$. Such a J_0 exists because $\dim e^\perp = N - 2$ is even. For $z \neq 0$ write $\hat{z} = z/\|z\|$. Given $\varepsilon \in \{+1, -1\}$, put*

$$e_\varepsilon = \varepsilon e, \quad U_\varepsilon = \{\hat{z} \in S(Z_N) : \hat{z} \neq -e_\varepsilon\}, \quad (54)$$

and, on U_ε ,

$$A_\varepsilon(\hat{z}) = \hat{z} \otimes e_\varepsilon^b - e_\varepsilon \otimes \hat{z}^b, \quad Q_\varepsilon(\hat{z}) = I + A_\varepsilon(\hat{z}) + \frac{A_\varepsilon(\hat{z})^2}{1 + \langle e_\varepsilon, \hat{z} \rangle}. \quad (55)$$

Then

$$C_z^{(\varepsilon)} = Q_\varepsilon(\hat{z}) J_0 Q_\varepsilon(\hat{z})^*, \quad r_z = \frac{2z^b}{\|z\|^2} \quad (56)$$

defines a smooth full-rank synthesis on the cone over U_ε . The two cones cover $Z_N \setminus \{0\}$ and therefore give an explicit two-chart atlas at every even order.

Proof. The Rodrigues identity in (55) gives $Q_\varepsilon^* Q_\varepsilon = I$ and $Q_\varepsilon e_\varepsilon = \hat{z}$. Since $\text{span}\{e_\varepsilon\} = \text{span}\{e\}$, conjugation yields $(C_z^{(\varepsilon)})^* = -C_z^{(\varepsilon)}$ and $\ker C_z^{(\varepsilon)} = \text{span}\{\hat{z}\} = \text{span}\{z\}$. Together with $r_z(z) = 2$, these are exactly the full-rank conditions (50). Finally $U_+ \cup U_- = S(Z_N)$. $\square$

Corollary 6.14 (Metric normalization and stable chart selection). *For either chart in Proposition 6.13,*

$$\boxed{(C_z^{(\varepsilon)})^2 = -(I - \hat{z} \otimes \hat{z}^b).} \quad (57)$$

Hence $C_z^{(\varepsilon)}$ vanishes on $\text{span}\{z\}$, restricts to an orthogonal complex structure on $z^\perp$, and has all nonzero singular values equal to one. Moreover,

$$\boxed{\|r_z\| = \frac{2}{\|z\|} = \min\{\|r\| : r \in Z_N^*, r(z) = 2\},} \quad (58)$$

and r_z is the unique minimizer. At evaluation time, choose $\varepsilon \in \{+1, -1\}$ so that $\varepsilon \langle e, \hat{z} \rangle \geq 0$; equivalently, take the sign of the inner product when it is nonzero and set $\varepsilon = +1$ when it vanishes. Then the Rodrigues denominator satisfies

$$1 + \langle e_\varepsilon, \hat{z} \rangle = 1 + |\langle e, \hat{z} \rangle| \in [1, 2]. \quad (59)$$

Proof. Since e is unit and J_0 is an orthogonal complex structure on $e^\perp$, $J_0^2 = -(I - e \otimes e^b)$. Orthogonal conjugation by Q_ε and $Q_\varepsilon e = \varepsilon \hat{z}$ give (57). For every covector r with $r(z) = 2$, Cauchy–Schwarz gives $2 \leq \|r\| \|z\|$, with equality exactly when r is a positive multiple of z^b . The constraint then fixes that multiple and gives (58). The final identity follows directly from the stated choice of sign. $\square$

Corollary 6.15 (Unavoidable radial blow-up). *Let N be even and normalize an admissible kernel as $\kappa = 1 + z$, with $z \in Z_N \setminus \{0\}$. In the fixed Euclidean normal-form coordinates, every compatible pairing having κ in its left radical has data (r_z, C_z) satisfying*

$$\boxed{\|r_z\| \geq \frac{2}{\|z\|}}. \quad (60)$$

Equality holds only for $r_z = 2z^\flat/\|z\|^2$. If $A_z = A_{r_z, C_z}$ is the composed Stokes matrix in (32), then

$$\boxed{\|A_z\|_F \geq \frac{2\sqrt{2}}{\|z\|}}. \quad (61)$$

If $M_z D = A_z$ in fixed orthonormal coefficient coordinates, then

$$\boxed{\|M_z\|_F \geq \frac{2\sqrt{2}}{\|D\|_2 \|z\|}}. \quad (62)$$

Consequently, no full-rank synthesis can remain bounded while approaching the deleted constant class $z = 0$, including in the exceptional orders $N = 2, 4, 8$. The numerical constants depend on the auxiliary metric and fixed coefficient coordinates, but the unboundedness and the lower-bound order $\|z\|^{-1}$ are invariant, up to multiplicative constants, under every fixed change of coordinates.

Proof. The general radical equation (35) gives $r_z(z) = 2$, so Cauchy–Schwarz yields (60) and its equality case. In the normal form (32), the vector r_z occurs in two disjoint blocks; hence

$$\|A_z\|_F^2 = 4 + 2\|r_z\|^2 + \|C_z\|_{\text{HS}}^2 \geq 2\|r_z\|^2,$$

which proves (61). Finally, $\|M_z D\|_F \leq \|M_z\|_F \|D\|_2$ gives (62). For the explicit normalized sections, $\|C_z\|_{\text{HS}}^2 = N - 2$ and $\|r_z\|^2 = 4/\|z\|^2$, so the divergence rate $\|z\|^{-1}$ is attained by A_z . Since D is fixed and surjective, the reconstruction $A \mapsto M$ defined by $MD = A$ is a linear isomorphism from the forms vanishing on $\ker D$ in the second slot to the mixed pairings, and both it and its inverse are bounded. Hence the same rate is attained by M_z , up to coordinate-dependent constants. Only the radial component is norm-minimal; the unit tangent normalization is a conditioning convention, not a minimum-norm claim for C_z . $\square$

Remark 6.16 (Algorithmic meaning). Every admissible kernel has a full-rank realization at every order. Proposition 6.13 turns this pointwise fact into a closed-form, optimization-free synthesis with two explicit charts: the formulas give (C_z, r_z) , followed by the fixed linear reconstruction $M_z D = A_{r_z, C_z}$. Corollary 6.14 gives a uniformly normalized tangent block and an antipode-safe chart selector. It does not give a uniformly bounded full-rank synthesis near the deleted constant line, where Corollary 6.15 forces $\|r_z\| \geq 2/\|z\|$. Theorem 6.9 says exactly when those charts can be replaced by one global smooth rule. The sign selector is piecewise smooth and may jump across the equator; in the nonexceptional even orders this is consistent with the obstruction to a global section. The obstruction belongs to the choice of gauge over the base, not to the existence of individual fibres.

7 Low-order models

7.1 The complete $N = 2$ family

Use the monomial bases $(1, x, x^2)$ of E_2 and $(1, x)$ of F_2 . Every compatible pairing is represented, after the normalization used below, by

$$M_t = \begin{pmatrix} 2 & 0 \\ 0 & \frac{2}{3} + \frac{t}{2} \\ \frac{2}{3} - t & 0 \end{pmatrix}. \quad (63)$$

Its kernel polynomial is

$$\kappa_t(x) = x^2 + \frac{t}{2} - \frac{1}{3}. \quad (64)$$

When

$$a^2 = \frac{1}{3} - \frac{t}{2} \in (0, 1),$$

the intrinsic nodes are $\{-a, a\}$. Thus the one-dimensional gauge family covers exactly the symmetric interior pairs, in agreement with (43).

At the nodes $(-a, a)$ the exact pairing has the matrix representation

$$H_a = \frac{1}{4a^2} \begin{pmatrix} 1 + a^2 & 3a^2 - 1 \\ 3a^2 - 1 & 1 + a^2 \end{pmatrix}. \quad (65)$$

It is diagonal if and only if $a^2 = 1/3$, equivalently $t = 0$. In that case $H_a = I_2$ and the nodes are $\pm 1/\sqrt{3}$. For every $0 < a < 1$, the eigenvalues of H_a are

$$1, \quad \frac{1 - a^2}{2a^2},$$

so the matrix is positive definite even when it is nonlocal. This example already separates positive full-matrix power evaluation from positive scalar quadrature.

7.2 A balanced asymmetric triple

For $N = 3$, the node condition is

$$x_1 + x_2 + x_3 + x_1x_2x_3 = 0. \quad (66)$$

Indeed, this is the expansion of (42). Given x_1, x_2 with $1 + x_1x_2 \neq 0$, one may set

$$x_3 = -\frac{x_1 + x_2}{1 + x_1x_2}.$$

The rational triple

$$\left(-\frac{19}{20}, \frac{2}{5}, \frac{55}{62}\right) \quad (67)$$

is admissible and asymmetric. Its ordinary interpolatory weights for Lebesgue integration are

$$\frac{51200}{92259}, \quad \frac{18950}{12231}, \quad -\frac{53816}{515967}.$$

The negative third weight shows that Stokes admissibility alone does not imply positivity of the ordinary quadrature associated with the same roots.

8 Nodal representation and the rigidity of locality

8.1 Every simple real kernel gives a full nodal matrix

Suppose β has full rank and its degree- N kernel polynomial has distinct real roots $X = (x_1, \dots, x_N)$. Define evaluation maps

$$V_E p = (p(x_1), \dots, p(x_N))^T, \quad V_F r = (r(x_1), \dots, r(x_N))^T.$$

Then $V_F : F_N \rightarrow \mathbb{R}^N$ is an isomorphism, while $\ker V_E = \text{span}\{\kappa_\beta\} = \ker \beta^\beta$.

Proposition 8.1 (Unique full nodal representation). *There exists a unique invertible matrix $H_\beta \in \mathbb{R}^{N \times N}$ such that*

$$\boxed{\beta(p, r) = (V_E p)^\top H_\beta (V_F r)}, \quad p \in E_N, \quad r \in F_N. \quad (68)$$

Proof. The pairing descends to a nondegenerate bilinear form on $(E_N / \text{span}\{\kappa_\beta\}) \times F_N$. Both factors are identified with $\mathbb{R}^N$ by V_E and V_F , so a unique invertible representing matrix exists. $\square$

The matrix H_β couples different nodes. A classical scalar quadrature corresponds to the exceptional case in which it is diagonal.

8.2 A polynomial spanning lemma

Lemma 8.2.

$$\text{span}\{(pq)' : p, q \in \mathbb{P}_N\} = \mathbb{P}_{2N-1}. \quad (69)$$

Proof. For $0 \leq k \leq 2N-1$, let $j = k+1$ and choose nonnegative integers $a, b \leq N$ with $a+b = j$. Then

$$x^k = \frac{1}{k+1} (x^a x^b)'.$$

Thus every monomial of degree at most $2N-1$ belongs to the displayed span. $\square$

Theorem 8.3 (Scalar moment rigidity). *Suppose a Stokes-compatible pairing has the product form*

$$\beta(p, r) = \mathcal{L}(pr) \quad (70)$$

for a linear functional $\mathcal{L}$ on $\mathbb{P}_{2N-1}$. Then

$$\mathcal{L}(f) = \int_{-1}^1 f(x) dx, \quad f \in \mathbb{P}_{2N-1}, \quad (71)$$

and hence $\beta = \beta_0$.

Proof. Stokes compatibility gives

$$\mathcal{L}((pq)') = [pq]_{-1}^1 = \int_{-1}^1 (pq)' dx.$$

Lemma 8.2 implies equality of the two functionals on all of $\mathbb{P}_{2N-1}$. Products pr , with $p \in \mathbb{P}_N$ and $r \in \mathbb{P}_{N-1}$, belong to this space, so the pairings coincide. $\square$

This theorem already rules out a nontrivial weighted moment interpretation inside the fixed unweighted Stokes class.

8.3 Gauss–Legendre is the unique local scalar gauge

Theorem 8.4 (Rigidity of local scalar realization). *Let $\beta : E_N \times F_N \rightarrow \mathbb{R}$ be full rank and Stokes-compatible. Suppose there exist N distinct real points x_i and nonzero scalars w_i such that*

$$\beta(p, r) = \sum_{i=1}^N w_i p(x_i) r(x_i), \quad p \in E_N, \quad r \in F_N. \quad (72)$$

Then:

- (i) *the rule $\sum_i w_i f(x_i)$ is exact on $\mathbb{P}_{2N-1}$;*
- (ii) *the node polynomial is proportional to the Legendre polynomial P_N ;*
- (iii) *the x_i are the Gauss–Legendre nodes and the w_i are the positive Gauss weights;*

(iv) $\beta = \beta_0$, equivalently $\omega = 0$.

Conversely, β_0 has the representation (72) at the Gauss–Legendre nodes and weights.

Proof. Define $\mathcal{Q}(f) = \sum_i w_i f(x_i)$. Applying Stokes to $p, q \in \mathbb{P}_N$ gives

$$\mathcal{Q}((pq)') = [pq]_{-1}^1 = \int_{-1}^1 (pq)' dx.$$

Lemma 8.2 proves exactness on $\mathbb{P}_{2N-1}$.

Let $\pi_X(x) = \prod_i (x - x_i)$. For every $r \in \mathbb{P}_{N-1}$,

$$\int_{-1}^1 \pi_X r dx = \sum_i w_i \pi_X(x_i) r(x_i) = 0.$$

Thus π_X is the degree- N polynomial orthogonal to $\mathbb{P}_{N-1}$ and is therefore proportional to P_N . The nodes are its roots. The weights are uniquely determined by exactness on $\mathbb{P}_{N-1}$. If ℓ_i are the cardinal polynomials, exactness applied to $\ell_i^2 \in \mathbb{P}_{2N-2}$ gives

$$w_i = \int_{-1}^1 \ell_i(x)^2 dx > 0.$$

Finally, $pr \in \mathbb{P}_{2N-1}$, so (72) equals $\int pr$ for every admissible pair. This is β_0 . The converse is ordinary Gaussian exactness. $\square$

Corollary 8.5. *For a full-rank pairing with a simple real degree- N kernel, the matrix H_β in (68) is diagonal if and only if $\omega = 0$. Hence every nonzero gauge is nonlocal in its intrinsic kernel-node chart.*

Proof. If H_β is diagonal, its diagonal entries are nonzero because H_β is invertible; Theorem 8.4 then gives $\beta = \beta_0$ and $\omega = 0$. Conversely, the Gauss–Legendre formula represents β_0 by its positive diagonal weight matrix. $\square$

The conclusion remains true when a nonzero gauge happens to retain P_N as its kernel. The nodes alone do not determine the pairing because the kernel map has positive-dimensional fibres for $N \geq 3$.

9 Minimum-norm solution of a prescribed-kernel constraint

The affine Stokes structure alone carries no norm, so it cannot select a minimum gauge. Here we use the distinguished L^2 reference to make such a selection. Let

$$E_N^0 = \left\{ p \in E_N : \int_{-1}^1 p(x) dx = 0 \right\}.$$

The L^2 product identifies $E_N/\mathbb{R}$ with E_N^0 and induces a Euclidean norm $\|\cdot\|_{\Lambda^2}$ on $\Lambda^2(E_N/\mathbb{R})^*$. Write $Q_N^{(0)} = E_N/\mathbb{R}$ for this L^2 quotient and denote its inner product by $\langle \cdot, \cdot \rangle_0$. Every two-form ω has a unique skew-adjoint representative $K_\omega \in \text{End}(Q_N^{(0)})$,

$$\omega(p, q) = \langle [p], K_\omega[q] \rangle_0, \quad \|\omega\|_{\Lambda^2}^2 = \frac{1}{2} \|K_\omega\|_{\text{HS}}^2. \quad (73)$$

Denote by $\Delta\beta_\omega$ the unique mixed pairing satisfying

$$\Delta\beta_\omega(p, q') = \omega(p, q). \quad (74)$$

For $\beta = \beta_0 + \Delta\beta_{\omega_\beta}$, we use the induced gauge norm

$$\|\beta - \beta_0\| := \|\omega_\beta\|_{\Lambda^2} = \frac{1}{\sqrt{2}}\|K_{\omega_\beta}\|_{\text{HS}}. \quad (75)$$

For an admissible nonconstant polynomial κ , define $u = [\kappa] \in Q_N^{(0)}$ and let $g \in Q_N^{(0)}$ be the Riesz representative of the functional

$$\langle g, [q] \rangle_0 = \int_{-1}^1 \kappa(x)q'(x) dx. \quad (76)$$

The endpoint condition gives

$$\langle g, u \rangle_0 = \int_{-1}^1 \kappa \kappa' dx = \frac{1}{2}(\kappa(1)^2 - \kappa(-1)^2) = 0. \quad (77)$$

For $v \in Q_N^{(0)}$, the notation $v^\flat$ denotes the Riesz covector $w \mapsto \langle v, w \rangle_0$.

Let P_N denote the Legendre polynomial normalized by $P_N(1) = 1$, and set

$$s = \kappa(1), \quad w = sP_N - \kappa. \quad (78)$$

Then $w(1) = w(-1) = 0$.

Lemma 9.1 (Explicit Riesz vector). *The vector in (76) is*

$$g = w' = (sP_N - \kappa)'. \quad (79)$$

Proof. For every $q \in E_N$, orthogonality of P_N to $q' \in F_N$ gives

$$q(1) - (-1)^N q(-1) = \int_{-1}^1 P_N'(x)q(x) dx. \quad (80)$$

Integration by parts and $\kappa(-1) = (-1)^N s$ therefore yield

$$\int_{-1}^1 \kappa q' = s(q(1) - (-1)^N q(-1)) - \int_{-1}^1 \kappa' q = \int_{-1}^1 w' q.$$

Moreover, $\int_{-1}^1 w' = w(1) - w(-1) = 0$, so $w' \in E_N^0$ is precisely the Riesz representative in the zero-mean realization of $Q_N^{(0)}$. $\square$

Let

$$\mathcal{V}_\kappa = \left\{ \omega \in \Lambda^2(E_N/\mathbb{R})^* : (\beta_0 + \Delta\beta_\omega)(\kappa, r) = 0 \text{ for every } r \in F_N \right\}. \quad (81)$$

The full-rank locus in $\mathcal{V}_\kappa$ is defined by the nonvanishing of a maximal minor, so it is a relative Zariski-open subset. It is nonempty by Theorem 6.1; since $\mathcal{V}_\kappa$ is an affine space and hence irreducible, that locus is dense. Thus $\mathcal{V}_\kappa$ is exactly the affine closure of the full-rank fibre with prescribed kernel line.

Theorem 9.2 (Closed formula for the minimum gauge). *The radical constraint $\omega \in \mathcal{V}_\kappa$ is equivalent to*

$$K_\omega u = g. \quad (82)$$

Its unique minimum-norm solution is

$$K_{\text{can}} = \frac{g \otimes u^\flat - u \otimes g^\flat}{\|u\|_0^2}, \quad \|\omega_{\text{can}}\|_{\Lambda^2} = \frac{\|g\|_0}{\|u\|_0}. \quad (83)$$

Every solution is uniquely of the form

$$K_\omega = K_{\text{can}} + L, \quad L^* = -L, \quad Lu = 0, \quad (84)$$

and the two terms are Hilbert–Schmidt orthogonal. In particular, the residual affine freedom has dimension $(N-1)(N-2)/2$. The construction depends only on the projective line $[\kappa]$ and on the chosen L^2 quotient metric.

Proof. For $r = q'$, the radical equation is

$$0 = \int \kappa q' + \omega(\kappa, q) = \langle g, [q] \rangle_0 + \langle u, K_\omega[q] \rangle_0 = \langle g - K_\omega u, [q] \rangle_0,$$

which proves (82). Orthogonality (77) shows that the operator in (83) is skew and sends u to g . The difference of any two solutions is a skew operator annihilating u , proving (84). A direct trace computation gives $\langle K_{\text{can}}, L \rangle_{\text{HS}} = 0$ whenever $Lu = 0$; hence Pythagoras proves minimality and uniqueness. The operator acts only on $\text{span}\{u, g\}$, so $\frac{1}{2}\|K_{\text{can}}\|_{\text{HS}}^2 = \|g\|_0^2/\|u\|_0^2$. Finally, L is an arbitrary skew operator on $u^\perp$, which gives the dimension. Both u and g scale with κ , so (83) is projectively invariant. $\square$

Let $\omega_{\text{can}}(\kappa)$ be the two-form represented by K_{can} and put

$$\beta_{\text{can}}(\kappa) = \beta_0 + \Delta\beta_{\omega_{\text{can}}(\kappa)}. \quad (85)$$

Theorem 9.3 (Global rank stratification of the canonical representative). *For every admissible nonconstant $\kappa \in E_N$,*

$$\boxed{\text{rank } \beta_{\text{can}}(\kappa) = \begin{cases} N, & \deg \kappa = N, \\ N-2, & \deg \kappa < N. \end{cases}} \quad (86)$$

Put

$$a_{\text{can}}(p, q) = \beta_{\text{can}}(\kappa)(p, q'), \quad T_{\text{can}}q = \iota(q') + K_{\text{can}}[q], \quad (87)$$

where $\iota : F_N \hookrightarrow E_N$ is the natural inclusion. If $\deg \kappa = N$, then

$$\text{rad}_R a_{\text{can}} = \text{span}\{1\}, \quad \text{rad}_L a_{\text{can}} = \text{rad}_L \beta_{\text{can}}(\kappa) = \text{span}\{\kappa\}. \quad (88)$$

If $\deg \kappa < N$, define the zero-mean representative and its centred primitive by

$$\bar{\kappa} = \frac{1}{2} \int_{-1}^1 \kappa(x) dx, \quad \kappa_0 = \kappa - \bar{\kappa}, \quad \eta(x) = \int_{-1}^x \kappa_0(t) dt. \quad (89)$$

Then, with $w = sP_N - \kappa$ as in (78),

$$\boxed{\text{rad}_R a_{\text{can}} = \text{span}\{1, w, \eta\}, \quad \text{rad}_L a_{\text{can}} = \text{span}\{\kappa, w, \eta\}.} \quad (90)$$

Thus the lower-degree strata have an exact rank defect two. They occur only for $N \geq 3$.

Proof. Identifying quotient classes with their zero-mean representatives gives

$$a_{\text{can}}(p, q) = \langle p, T_{\text{can}}q \rangle_{L^2}.$$

The L^2 form on E_N is nondegenerate, and $d : E_N \rightarrow F_N$ is onto; hence

$$\text{rank } \beta_{\text{can}}(\kappa) = \text{rank } a_{\text{can}} = \text{rank } T_{\text{can}}. \quad (91)$$

Assume first that $\deg \kappa < N$. Orthogonality to P_N gives

$$\langle u, [w] \rangle_0 = -\|u\|_0^2, \quad \langle g, [w] \rangle_0 = \int_{-1}^1 w' w dx = 0. \quad (92)$$

Formula (83) therefore yields $K_{\text{can}}[w] = -g$, and thus $T_{\text{can}}w = 0$. Since κ_0 has zero mean, the polynomial η in (89) vanishes at both endpoints. Moreover,

$$\langle u, [\eta] \rangle_0 = \int_{-1}^1 \eta' \eta \, dx = 0, \quad \langle g, [\eta] \rangle_0 = \int_{-1}^1 w' \eta \, dx = - \int_{-1}^1 w \kappa_0 \, dx = \|u\|_0^2. \quad (93)$$

Consequently $K_{\text{can}}[\eta] = -u$ and $T_{\text{can}}\eta = \eta' - \kappa_0 = 0$. Also $T_{\text{can}}1 = 0$.

The polynomials $1, w, \eta$ are linearly independent. Indeed, evaluating a possible relation at either endpoint first removes its constant term. If $\deg \kappa \leq N - 2$, then w and η have different degrees. If $\deg \kappa = N - 1$, proportionality of w and η would imply proportionality of the nonzero vectors $g = w'$ and $u = \kappa_0$, contrary to $\langle g, u \rangle_0 = 0$. Hence

$$\text{rank } T_{\text{can}} \leq N - 2. \quad (94)$$

On the other hand, $\iota d : E_N \rightarrow E_N$ has rank N , while K_{can} has rank two because u and g are nonzero and orthogonal. The rank inequality gives

$$\text{rank } T_{\text{can}} = \text{rank}(\iota d + K_{\text{can}}) \geq \text{rank}(\iota d) - \text{rank } K_{\text{can}} = N - 2. \quad (95)$$

Equations (91)–(95) prove the lower branch of (86) and the right-radical identity in (90).

Both w and η vanish at the endpoints. Since they lie in the right radical, the Stokes identity puts them in the left radical as well. The polynomial κ is in the left radical by construction and is independent of w, η because its endpoint value is nonzero. The left nullity is three, so these vectors give the second identity in (90). For $N = 1$ there is no nonconstant lower-degree polynomial; for $N = 2$, the even endpoint condition forces every affine admissible polynomial to be constant. Hence the lower-degree branch can occur only for $N \geq 3$.

Assume now that $\deg \kappa = N$ and let p belong to the left radical. Since both p and κ annihilate F_N ,

$$0 = b(\kappa, p) = s(p(1) - (-1)^N p(-1)).$$

Put $t = p(1)$ and $h = tP_N - p$. Then $h(1) = h(-1) = 0$, and the calculation of Lemma 9.1, now applied to p , shows that its Riesz vector is h' . The radical equation is therefore

$$h' = K_{\text{can}}[p] = \frac{g\langle u, [p] \rangle_0 - u\langle g, [p] \rangle_0}{\|u\|_0^2}. \quad (96)$$

The polynomials h' and g have degree at most $N - 1$, whereas the zero-mean representative of u has degree exactly N . Comparing the degree- N component in (96) gives $\langle g, [p] \rangle_0 = 0$. With

$$a = \frac{\langle u, [p] \rangle_0}{\|u\|_0^2},$$

we obtain $h' = ag$. The polynomials h and aw have the same derivative and both vanish at the endpoints; hence $h = aw$. Consequently,

$$p = a\kappa + (t - as)P_N. \quad (97)$$

Taking the inner product with u and using the definition of a gives

$$(t - as)\langle u, [P_N] \rangle_0 = 0.$$

Because κ has exact degree N , its coefficient along P_N is nonzero, so $\langle u, [P_N] \rangle_0 \neq 0$. Thus $t = as$ and $p = a\kappa$. The left radical is exactly one dimensional, which is equivalent to full rank. The right radical of a_{can} contains the constants and, by rank-nullity, is exactly $\text{span}\{1\}$. $\square$

Corollary 9.4 (Attainment of the minimum gauge in a full-rank fibre). *Let $\mathcal{F}_\kappa^{\text{fr}}$ be the full-rank fibre over an admissible kernel line $[\kappa]$. Then*

$$\inf_{\beta \in \mathcal{F}_\kappa^{\text{fr}}} \|\beta - \beta_0\| = \|\beta_{\text{can}}(\kappa) - \beta_0\|. \quad (98)$$

The infimum is attained if and only if $\deg \kappa = N$; in that case its unique minimizer is $\beta_{\text{can}}(\kappa)$. If $\deg \kappa < N$, the infimum is not attained.

Proof. The prescribed-kernel constraint is affine and closed, and its unique minimum-norm element is $\beta_{\text{can}}(\kappa)$. Full rank is an open dense condition in every nonempty fibre, so the infimum over the full-rank part equals the minimum over its closure. By Theorem 9.3, the canonical representative is full rank exactly when $\deg \kappa = N$. Strict convexity of the squared norm gives uniqueness in that case; in lower degree the unique minimizer lies outside the full-rank locus, so the common infimum cannot be attained there. $\square$

Corollary 9.5 (Rectangular right radical). *For a mixed pairing write*

$$\text{rad}_R \beta = \{r \in F_N : \beta(p, r) = 0 \text{ for every } p \in E_N\}.$$

The canonical representative satisfies

$$\text{rad}_R \beta_{\text{can}}(\kappa) = \begin{cases} \{0\}, & \deg \kappa = N, \\ \text{span}\{g, \kappa_0\}, & \deg \kappa < N, \end{cases} \quad (99)$$

where $g = w'$ and $\kappa_0 = \eta'$ are nonzero and L^2 -orthogonal in the second case. Thus the rectangular left and right nullities are respectively $(1, 0)$ in exact degree and $(3, 2)$ in lower degree.

Proof. Surjectivity of $d : E_N \rightarrow F_N$ identifies F_N with $E_N/\mathbb{R}$. The right radical of β_{can} is therefore the derivative of the right radical of a_{can} . Equations (88) and (90) give respectively $\{0\}$ and $\text{span}\{w', \eta'\}$. Nonvanishing and orthogonality follow from the proof of Theorem 9.3. $\square$

Lemma 9.6 (Rectangular singular-value opening). *Let $M(t) : X \rightarrow Y$ be a smooth family between finite-dimensional Hilbert spaces with $\dim Y \geq \dim X = n$, and suppose $\text{rank } M(0) = n - q$. Put*

$$R = \ker M(0), \quad L = \ker M(0)^*, \quad G = \Pi_L M'(0)|_R : R \rightarrow L. \quad (100)$$

If G is injective, then there are $c, C, \delta > 0$ such that, for $0 < |t| < \delta$,

$$c|t| \leq \sigma_j(M(t)) \leq C|t|, \quad j = n - q + 1, \dots, n, \quad (101)$$

where the singular values are the eigenvalues of $(M(t)^ M(t))^{1/2}$, counted with multiplicity and ordered decreasingly. The other $n - q$ singular values remain uniformly separated from zero.*

Proof. Use the orthogonal decompositions $X = R^\perp \oplus R$ and $Y = \text{im } M(0) \oplus L$. In the corresponding blocks,

$$M(t) = \begin{pmatrix} A(t) & B(t) \\ C(t) & D(t) \end{pmatrix}, \quad A(0) : R^\perp \rightarrow \text{im } M(0) \text{ is an isomorphism,}$$

and $B(0) = C(0) = D(0) = 0$. For small t , bounded invertible block eliminations reduce $M(t)$ to $\text{diag}(A(t), S(t))$, where

$$S(t) = D(t) - C(t)A(t)^{-1}B(t) = tG + O(t^2).$$

Injectivity of G gives two-sided linear singular-value bounds for $S(t)$. These bounds survive the eliminations because, for invertible P, Q and every admissible index j ,

$$\frac{\sigma_j(B)}{\|P^{-1}\| \|Q^{-1}\|} \leq \sigma_j(PBQ) \leq \|P\| \|Q\| \sigma_j(B). \quad (102)$$

The singular values belonging to $A(t)$ remain separated from zero by continuity. $\square$

Theorem 9.7 (Transverse opening of the lower-degree defect). *Let $\kappa_\star$ be an admissible lower-degree kernel, set*

$$s = \kappa_\star(1), \quad u_\star = \kappa_\star - \frac{1}{2} \int_{-1}^1 \kappa_\star(x) dx, \quad A = \|u_\star\|_{L^2}^2, \quad B = \|P_N\|_{L^2}^2, \quad (103)$$

and define $\kappa_t = \kappa_\star + tP_N$. For $t \neq -s$, this remains endpoint-admissible. Let a_t be the composed canonical form associated with κ_t , and set $T_t q = \iota(q') + K_{\text{can}}(\kappa_t)[q]$. Let

$$w = sP_N - \kappa_\star, \quad g = w', \quad \eta(x) = \int_{-1}^x u_\star(\tau) d\tau. \quad (104)$$

Then the restriction of a_t to the ordered basis (w, η) is exactly

$$[a_t]_{(w, \eta)} = \frac{ABt(s+t)}{A+Bt^2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (105)$$

In particular,

$$\left. \frac{d}{dt} [a_t]_{(w, \eta)} \right|_{t=0} = sB \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (106)$$

so the rank-two defect opens transversely. With respect to any fixed Hilbert structures, equivalently in fixed orthonormal bases, the two vanishing singular values $\sigma_{N-1}(\beta_t)$ and $\sigma_N(\beta_t)$ of the rectangular pairing satisfy

$$c|t| \leq \sigma_j(\beta_t) \leq C|t|, \quad j = N-1, N, \quad (107)$$

for some $c, C > 0$ and all sufficiently small $t \neq 0$, while the remaining singular values stay bounded away from zero. Consequently $\text{cond}(\beta_t) = \sigma_1(\beta_t)/\sigma_N(\beta_t)$ is of order $|t|^{-1}$.

Proof. Since $u_\star$, g , and w' have degree below N , Legendre orthogonality gives

$$\langle u_\star, P_N \rangle = \langle g, P_N \rangle = 0, \quad \|u_\star + tP_N\|^2 = A + Bt^2. \quad (108)$$

Moreover,

$$\langle u_\star, w \rangle = -A, \quad \langle P_N, w \rangle = sB, \quad \langle g, w \rangle = 0, \quad \langle g, \eta \rangle = A. \quad (109)$$

The Legendre defect associated with κ_t is still w , hence its Riesz vector is still g . Substitution into (83) yields

$$T_t w = \frac{Bt(s+t)}{A+Bt^2} g. \quad (110)$$

Because w and η vanish at both endpoints, the restriction of a_t to their span is skew. Equations (109)–(110) give $a_t(\eta, w) = ABt(s+t)/(A+Bt^2)$, proving (105) and (106). At $t = 0$, the corresponding left null vectors are w, η and the right null vectors of the mixed pairing are $g, \eta' = u_\star$ by Corollary 9.5. The derivative block is nondegenerate because $sB \neq 0$. Therefore the crossing map in Lemma 9.6 is injective on the two-dimensional right kernel. The lemma gives (107) and the claimed condition-number estimate. $\square$

Remark 9.8 (Pairing section versus nodal section). The projectively invariant map $[\kappa] \mapsto \beta_{\text{can}}(\kappa)$ is a global canonical pairing section on the entire admissible degree- N kernel stratum, including classes with multiple or nonreal roots. A nodal matrix requires N distinct roots, and an ordered interior node configuration requires those roots to be real and simple. Thus the canonical interior nodal section below is the restriction of the pairing section to the simple interior real-root stratum.

At the Legendre line, $w = 0$ and $g = 0$, so the canonical gauge is the zero gauge.

For an admissible ordered interior node set $X = (x_1, \dots, x_N)$, put $\pi_X(x) = \prod_i (x - x_i)$, let $\omega_{\text{can}}(X)$ be the two-form represented by (83) with $\kappa = \pi_X$, and define $\beta_{\text{can}}(X) = \beta_0 + \Delta\beta_{\omega_{\text{can}}(X)}$.

Corollary 9.9 (Global canonical nodal section and local hyperbolic stability). *For every simple admissible interior configuration X , the pairing $\beta_{\text{can}}(X)$ has full rank and its intrinsic nodal matrix $H_{\text{can}}(X)$ is well defined. Let $X_G = (\xi_1, \dots, \xi_N)$ be the ordered Gauss–Legendre configuration and put $y_i = \text{artanh } x_i$, $y_i^G = \text{artanh } \xi_i$. There exist a neighbourhood U of X_G in the admissible configuration manifold and a constant C_U such that, for every $X \in U$,*

$$\|\omega_{\text{can}}(X)\|_{\Lambda^2} + \|\text{off } H_{\text{can}}(X)\|_{\text{F}} \leq C_U \|y - y^G\|_2. \quad (111)$$

Here $\text{off } H = H - \text{diag}(H_{11}, \dots, H_{NN})$.

Proof. In fixed polynomial coordinates, the coefficients of π_X , and hence the vectors $u(X)$ and $g(X)$ in (76), depend smoothly on X . Since $u(X) \neq 0$, formula (83) makes $\omega_{\text{can}}(X)$ smooth on the degree- N admissible stratum. At X_G , the Legendre polynomial is orthogonal to F_N , so $g(X_G) = 0$ and $\omega_{\text{can}}(X_G) = 0$.

Theorem 9.3 gives full rank because $\deg \pi_X = N$. On the simple-node stratum, the evaluation isomorphisms in Proposition 8.1 depend smoothly on X ; therefore $H_{\text{can}}(X)$ is smooth. At X_G it is the diagonal matrix of positive Gauss weights. Thus both maps on the left of (111) vanish at X_G and are locally Lipschitz. Finally $x \mapsto \text{artanh } x$ is a smooth local diffeomorphism, so rapidity and Euclidean coordinates are locally bi-Lipschitz. $\square$

The estimate cannot be extended uniformly to the whole admissible chamber.

Proposition 9.10 (Failure of a global hyperbolic bound). *There is no constant C such that*

$$\|\text{off } H_{\text{can}}(X)\|_{\text{F}} \leq C \|\text{artanh } X - \text{artanh } X_G\|_2 \quad (112)$$

for every simple admissible interior configuration, even when $N = 2$.

Proof. For $N = 2$ every fibre has dimension zero, so the pairing at $X(a) = (-a, a)$ is forced and its nodal matrix is (65). Hence

$$\|\text{off } H_a\|_{\text{F}} = \frac{\sqrt{2} |3a^2 - 1|}{4a^2} \longrightarrow \infty \quad (a \downarrow 0).$$

On the other hand, with $c = \text{artanh}(1/\sqrt{3})$,

$$\|\text{artanh } X(a) - \text{artanh } X_G\|_2 = \sqrt{2} |\text{artanh } a - c| \longrightarrow \sqrt{2} c < \infty.$$

Thus (112) is impossible. $\square$

The degeneration $a \downarrow 0$ is a node collision at finite hyperbolic distance. Any global stability estimate must therefore contain an additional factor measuring separation, the nodal discriminant, or the condition number of the evaluation map. Hyperbolic distance controls displacement from Gauss but cannot by itself control collision geometry.

10 Ordinary quadrature attached only to the roots

Every set of distinct real nodes admits an interpolatory rule for Lebesgue integration, independently of the pairing gauge. Let ℓ_i be the Lagrange basis and define

$$\hat{w}_i = \int_{-1}^1 \ell_i(x) dx, \quad Q_X(f) = \sum_i \hat{w}_i f(x_i). \quad (113)$$

The rule is exact on $\mathbb{P}_{N-1}$, but it need not represent β and its weights need not be positive.

Proposition 10.1 (Degree of the interpolatory rule). *Let $\kappa_X = \prod_i (x - x_i)$ and let $s \in \{0, \dots, N\}$ be maximal such that*

$$\int_{-1}^1 \kappa_X(x) q(x) dx = 0, \quad q \in \mathbb{P}_{s-1}, \quad (114)$$

where the condition is empty for $s = 0$. Then Q_X has degree of exactness $N + s - 1$. If κ_X is Stokes-admissible but not proportional to P_N , its degree is at most $2N - 3$.

Proof. For $f \in \mathbb{P}_{N+s-1}$, polynomial division gives $f = \kappa_X q + r$ with $q \in \mathbb{P}_{s-1}$ and $r \in \mathbb{P}_{N-1}$. Since $Q_X(\kappa_X q) = 0$ and $Q_X(r) = \int r$, exactness is equivalent to (114). Maximality gives the degree when $s < N$. If $s = N$, the rule is exact through degree $2N - 1$, while $Q_X(\kappa_X^2) = 0 \neq \int_{-1}^1 \kappa_X^2$ shows that it is not exact in degree $2N$.

If the degree were at least $2N - 2$, then $\kappa_X \perp \mathbb{P}_{N-2}$, so $\kappa_X = aP_N + bP_{N-1}$. With the normalization $P_j(1) = 1$, Stokes admissibility gives

$$a + b = (-1)^N (a(-1)^N + b(-1)^{N-1}) = a - b,$$

hence $b = 0$. □

Thus there are three distinct statements:

1. the roots of κ_β are intrinsic to the gauged pairing;
2. those roots always support a full matrix representation of β ;
3. they support a scalar quadrature representation of β only in the zero gauge.

11 Orthogonal-polynomial families and the endpoint obstruction

Every parity-symmetric polynomial family satisfies the endpoint condition degree by degree. Consequently, Legendre, Chebyshev, Gegenbauer, symmetric Jacobi, and orthogonal polynomials for an even positive measure can each be realized as kernel polynomials of suitable gauges, provided their endpoint values are nonzero. This statement concerns the kernel line only; it does not identify the gauged pairing with the positive measure that originally defined the family.

The endpoint constraint becomes especially informative for a coherent orthogonal hierarchy.

Theorem 11.1 (Centered Favard recurrence). *Let $(\kappa_n)_{n \geq 0}$ be monic orthogonal polynomials for a positive measure supported on $[-1, 1]$, with recurrence*

$$\kappa_{n+1}(x) = (x - a_n)\kappa_n(x) - b_n\kappa_{n-1}(x), \quad b_n > 0. \quad (115)$$

Assume for every n that

$$\kappa_n(1) = (-1)^n \kappa_n(-1) \neq 0. \quad (116)$$

Then

$$a_n = 0 \quad \text{for every } n. \quad (117)$$

Hence the Jacobi matrix is centred and the orthogonality measure is symmetric.

Proof. Write $s_n = \kappa_n(1)$. Evaluating (115) at 1 gives

$$s_{n+1} = (1 - a_n)s_n - b_ns_{n-1}.$$

Evaluating at -1 and using (116) gives

$$s_{n+1} = (1 + a_n)s_n - b_ns_{n-1}.$$

Since $s_n \neq 0$, comparison yields $a_n = 0$. A compactly supported positive measure is determined by its Jacobi coefficients; vanishing diagonal coefficients give invariance under $x \mapsto -x$. □

The theorem explains why the fixed unweighted Stokes boundary is naturally compatible with centred classical hierarchies. General Jacobi families with unequal parameters require a weighted or otherwise modified Stokes geometry.

12 Relation with SBP and open Dirac structures

12.1 The skew freedom in SBP

The algebraic SBP identity is

$$Q + Q^\top = B. \quad (118)$$

Its general solution is $Q = \frac{1}{2}B + S$ with $S^\top = -S$. In the rectangular construction, Q is replaced by

$$A = MD,$$

and the derivative kernel additionally imposes $A(\cdot, 1) = 0$. Theorem 4.1 is therefore the exact quotient-aware version of the familiar skew freedom in SBP.

The containment of square SBP in the rectangular construction can be stated without a choice of polynomial basis.

Proposition 12.1 (Symmetric rectangularization of SBP). *Let $D \in \mathbb{R}^{n \times n}$ satisfy*

$$HD + D^\top H = B, \quad H = H^\top, \quad (119)$$

and assume $\ker D = \text{span}\{\mathbf{1}\}$. Put $F = \text{im } D$, let $J : F \hookrightarrow \mathbb{R}^n$ be the inclusion, and let $D : \mathbb{R}^n \rightarrow F$ be the corestriction of D , so that $D = JD$. Then

$$M := HJ : F \longrightarrow (\mathbb{R}^n)^* \quad (120)$$

satisfies the rectangular Stokes identity

$$MD + D^\top M^\top = B. \quad (121)$$

If H is nonsingular, then M has full column rank. Conversely, a rectangular factorization $D = JD$, $M = HJ$ with symmetric H turns (121) into the square SBP identity (119).

Proof. Substitution of $D = JD$ and $M = HJ$ gives

$$MD + D^\top M^\top = HJD + D^\top J^\top H = HD + D^\top H = B.$$

The inclusion J has full column rank, so HJ does as well when H is nonsingular. The same identities read in reverse prove the converse. $\square$

For nodal polynomial SBP, F is the sampled derivative space and D is the exact rectangular derivative after choosing coordinates in that space. For FSBP, the same argument applies with the sampled derivative image of the chosen function space, whenever constants form the derivative kernel. Proposition 12.1 is therefore an algebraic containment statement: such a square identity is a symmetric factorization of the mixed Stokes relation, with the additional restriction $M = HJ$. It is a metric factorization only when H is positive definite.

The phrase *dual-pairing SBP* is used elsewhere for paired forward and backward difference operators acting on the same discrete grid [27]. That notion should not be identified terminologically with the present mixed pairing $E \times F \rightarrow \mathbb{R}$, whose two factors have different dimensions and are connected by the surjective derivative. The frameworks share a power-balance identity but organize the discrete variables differently.

In the usual unreduced nodal coordinates, an SBP operator is written with a symmetric positive norm matrix P :

$$PD_h + D_h^\top P = R^\top \Sigma R.$$

Generalized SBP links positive norm matrices to quadrature [15, 10]; FSBP carries the same principle to arbitrary function spaces [12]. The present classification concerns the prior rectangular level, where no symmetric positive norm has yet been selected.

12.2 Boundary completion

Let $R : E_N \rightarrow \mathbb{R}^s$ be a collection of boundary traces. The augmented power map

$$p \mapsto (\beta^b p, Rp)$$

is injective if and only if

$$\ker \beta^b \cap \ker R = \{0\}. \quad (122)$$

For a full-rank pairing this is equivalent to $R\kappa_\beta \neq 0$. The endpoint condition (37) ensures that either endpoint trace completes the kernel direction on the degree-regular full-rank stratum. This proves the injective boundary-completion mechanism used in the 2012 construction. It does not, by itself, prove maximal isotropy of a complete open Dirac relation: that statement also requires specifying the full flow–effort relation, its boundary-port pairing, and the ambient symmetric form.

Gauge-dependent kernel nodes and arbitrary collocation nodes must not be conflated. Any unisolvent grid may represent the flow polynomial. The zeros of κ_β instead describe a quotient chart intrinsic to the chosen pairing, and Corollary 8.5 shows that this chart is generally matrix-weighted rather than diagonal.

13 Infinite-dimensional interpretation

13.1 The continuous affine class

The finite-dimensional classification has an exact functional analogue. Let

$$V = H^1(-1, 1), \quad W = L^2(-1, 1), \quad d = \partial_x.$$

The derivative induces a Banach-space isomorphism

$$\bar{d} : V/\mathbb{R} \rightarrow W.$$

Theorem 13.1 (Continuous Stokes pairing gauge). *The continuous bilinear pairings $\beta : V \times W \rightarrow \mathbb{R}$ satisfying*

$$\beta(u, v') + \beta(v, u') = [uv]_{-1}^1 \quad (123)$$

form an affine space with origin $\beta_0(u, f) = \int_{-1}^1 uf$ and model space

$$\Lambda_{\text{cont}}^2(V/\mathbb{R})^*.$$

Explicitly, for every continuous skew form Ω on $V/\mathbb{R}$,

$$\beta_\Omega(u, f) = \int_{-1}^1 u(x)f(x) dx + \Omega([u], \bar{d}^{-1}f) \quad (124)$$

satisfies (123), and every compatible continuous pairing has this form.

Proof. If β_1 and β_0 are compatible, define $\Omega(u, v) = (\beta_1 - \beta_0)(u, v')$. Subtraction of the Stokes identities makes Ω skew, and $v' = 0$ for constants, so it descends to $V/\mathbb{R}$. Conversely, (124) gives

$$\beta_\Omega(u, v') + \beta_\Omega(v, u') = \int (uv)' + \Omega([u], [v]) + \Omega([v], [u]) = [uv]_{-1}^1.$$

Uniqueness follows from surjectivity of d . □

Equip $Q^{(1)} = H^1(-1, 1)/\mathbb{R}$ with $\|[u]\|_1 = \|u'\|_{L^2}$. Then $\bar{d} : Q^{(1)} \rightarrow W$ is an isometry. This derivative metric is distinct from the L^2 quotient metric on $Q_N^{(0)}$ used in Section 9; the superscripts record that distinction. Riesz representation therefore sharpens Theorem 13.1 as follows.

Proposition 13.2 (Operator form and coherent discrete extension). *Every continuous gauge Ω has a unique representation*

$$\Omega([u], [v]) = \langle u', Sv' \rangle_{L^2}, \quad S^* = -S \in \mathcal{L}(L^2), \quad (125)$$

and the compatible pairing is

$$\beta_S(u, f) = \int_{-1}^1 u f \, dx + \langle u', Sf \rangle_{L^2}. \quad (126)$$

Let $Q_N^{(1)} = \mathbb{P}_N / \mathbb{R} \subset Q^{(1)}$ and let $\omega_N \in \Lambda^2(Q_N^{(1)})^*$. There exists a continuous two-form Ω on $Q^{(1)}$ with $\Omega|_{Q_N^{(1)} \times Q_N^{(1)}} = \omega_N$ for every N if and only if

$$\omega_{N+1}|_{Q_N^{(1)} \times Q_N^{(1)}} = \omega_N, \quad (127)$$

$$\sup_N \|\omega_N\|_{\text{op}} < \infty, \quad (128)$$

where the operator norm is induced by $\|\cdot\|_1$. The extension, and hence S , is unique.

Proof. The isometry $\bar{d}$ transports continuous skew forms on $Q^{(1)}$ bijectively to bounded skew forms on L^2 ; Riesz representation gives the unique skew-adjoint operator S and (126). Necessity of (127)–(128) is immediate. Conversely, restriction compatibility defines a skew form on the increasing union $\bigcup_N Q_N^{(1)}$, and the uniform bound makes it continuous there. Polynomials are dense in H^1 in the derivative norm modulo constants, so the form extends uniquely to $Q^{(1)} \times Q^{(1)}$. $\square$

Thus fixed-degree classification alone does not guarantee a continuum limit: restriction coherence and uniform operator control are the exact additional requirements.

13.2 Dirac B -transform

A two-form Ω defines

$$\tau_\Omega(v, e) = (v, e + \iota_v \Omega)$$

on $T(V/\mathbb{R}) \oplus T^*(V/\mathbb{R})$. Because Ω is skew, τ_Ω preserves the canonical symmetric power pairing. Since the form is constant, it is closed. It therefore has the algebraic form of a gauge transformation by a closed two-form [8]. This identifies the interior linear transformation. To conclude that it transforms a particular open boundary-port Dirac structure, one must additionally verify its domain, boundary trace relation, and maximal isotropy; those analytical data are not part of the finite-dimensional affine theorem. An analytical Hilbert-space treatment of these domain and boundary-duality questions is given in [7].

This two-form gauge is related to, but distinct from, the gauge reduction of PHS potentials. In the latter, a differential form ρ is replaced by $\rho + d\alpha$ and the physical variable $d\rho$ is unchanged; Stokes–Dirac and simplicial Dirac reductions of this type are studied in [23]. That reduction creates the quotient on which the present two-form lives; it does not itself vary the mixed pairing.

13.3 The Le Gorrec–Zwart–Maschke skew freedom

Le Gorrec, Zwart, and Maschke consider formally skew differential operators of the form

$$J = \sum_{i=0}^r P^{(i)} \partial_x^i, \quad P^{(i)} = (-1)^{i+1} P^{(i)\top}. \quad (129)$$

In particular, $P^{(0)\top} = -P^{(0)}$. Consequently,

$$\langle e_1, P^{(0)} e_2 \rangle_{L^2} + \langle e_2, P^{(0)} e_1 \rangle_{L^2} = 0, \quad (130)$$

and $P^{(0)}$ disappears from the Green boundary form. Indeed, the boundary matrix Q in their Theorem 3.1 depends only on $P^{(1)}, \dots, P^{(r)}$ [20]. This is the same power-invisibility principle that produces the present affine gauge: Stokes fixes the symmetric part of an interior bilinear expression and leaves its antisymmetric part undetermined.

The precise finite-dimensional relation requires the derivative quotient. Endow E_N with the unweighted L^2 inner product and regard $dq = q'$ as an element of E_N through the natural inclusion $F_N \subset E_N$.

Proposition 13.3 (Riesz representation of the pairing gauge). *The Riesz map defines a linear isomorphism*

$$\Lambda^2(E_N/\mathbb{R})^* \simeq \mathfrak{k}_N := \{K_N \in \text{End}(E_N) : K_N^* = -K_N, K_N 1 = 0\}. \quad (131)$$

In particular,

$$\dim \mathfrak{k}_N = \frac{N(N-1)}{2}. \quad (132)$$

More explicitly, every pairing gauge has a unique representation

$$\omega(p, q) = \langle p, K_N q \rangle_{L^2}, \quad (133)$$

and hence

$$a_\beta(p, q) = \langle p, (d + K_N)q \rangle_{L^2}. \quad (134)$$

Moreover, there is a unique map $L_N : F_N \rightarrow E_N$ such that

$$\boxed{K_N = L_N d}. \quad (135)$$

Thus the pairing gauge is exactly a skew-adjoint, zero-power correction to the discrete interconnection, subject to annihilating the derivative kernel.

Proof. For fixed q , the functional $p \mapsto \omega(p, q)$ has a unique L^2 Riesz representative, which defines $K_N q$ by (133). Skew-symmetry of ω is equivalent to $K_N^* = -K_N$. Since $\omega(p, 1) = 0$ for all p , nondegeneracy of L^2 gives $K_N 1 = 0$.

Conversely, if $K_N^* = -K_N$ and $K_N 1 = 0$, then (133) is skew and vanishes whenever either argument is constant; it therefore defines a two-form on $E_N/\mathbb{R}$. This proves the isomorphism and (132), since $\dim(E_N/\mathbb{R}) = N$. Finally, if $dq_1 = dq_2$, then $q_1 - q_2$ is constant and $K_N q_1 = K_N q_2$. Hence

$$L_N(dq) := K_N q$$

is well defined. Surjectivity of $d : E_N \rightarrow F_N$ proves existence and uniqueness of L_N , and gives (135). $\square$

Proposition 13.3 justifies calling the pairing gauge a *relative $P^{(0)}$ freedom*: after the L^2 identification, it is algebraically an antisymmetric addition to the interconnection operator, exactly as in (129). The qualifier “relative” is essential. The rectangular factorization imposes $K_N|_{\ker d} = 0$, a condition absent from a general $P^{(0)}$ term.

Corollary 13.4 (Locality obstruction). *Consider the componentwise derivative on $E_N \otimes \mathbb{R}^s$, whose kernel contains every constant vector field. Suppose an operator in the analogue of $\mathfrak{k}_N$ is the restriction of a pointwise multiplication operator*

$$(K_N q)(x) = P^{(0)}(x)q(x), \quad P^{(0)}(x)^\top = -P^{(0)}(x). \quad (136)$$

Then $P^{(0)}(x) = 0$ almost everywhere. In particular, every nonzero pairing gauge in the scalar polynomial setting fails to be a standard local order-zero multiplication term when transferred to the interconnection operator; its nodal or modal realization is generically nonlocal.

Proof. For every $c \in \mathbb{R}^s$, the constant function $q(x) = c$ belongs to the derivative kernel. Therefore $0 = K_N c = P^{(0)}(x)c$ almost everywhere. Since this holds for every c , $P^{(0)}(x) = 0$ almost everywhere. For $s = 1$, the conclusion also follows directly because a real skew-symmetric 1×1 matrix is zero. $\square$

There is a second freedom in the construction of Le Gorrec, Zwart, and Maschke. If R_{ext} factorizes their boundary form, every other admissible factorization is

$$R = UR_{\text{ext}}, \quad U^\top \Sigma U = \Sigma. \quad (137)$$

This is a power-preserving change of boundary-port coordinates. It leaves the differential operator and its Green form unchanged and is therefore distinct from the present pairing gauge, which changes β , its radical, and generally the roots of κ_β .

We conclude that the three notions require a strict distinction. The $P^{(0)}$ term expresses local component-skew freedom in the continuum operator; K_N expresses spatial-mode skew freedom compatible with the derivative quotient; and U in (137) expresses only a boundary-coordinate freedom. They share preservation of power, but they do not act on the same object. A nonzero sequence (K_N) can represent a genuine continuum operator only under additional coherence, locality, and uniform-boundedness hypotheses.

13.4 Why a change of weight is a different geometry

Let w be smooth and strictly positive on the closed interval, and define

$$g_w(u, v) = \int_{-1}^1 wuv.$$

The covariant first-order operator

$$\nabla_w = \partial_x + \frac{1}{2} \frac{w'}{w} \quad (138)$$

satisfies

$$g_w(u, \nabla_w v) + g_w(v, \nabla_w u) = [wuv]_{-1}^1. \quad (139)$$

Thus changing weight modifies the symmetric metric, the differential operator, and the boundary form together. It is conjugate to the unweighted derivative by multiplication with $\sqrt{w}$, but it does not belong to the affine class with d and b fixed.

More generally, for a coefficient σ define

$$\nabla_{\sigma, w} = \sigma \partial_x + \frac{1}{2} \frac{(\sigma w)'}{w}.$$

Then

$$g_w(u, \nabla_{\sigma, w} v) + g_w(v, \nabla_{\sigma, w} u) = [\sigma wuv]_{-1}^1. \quad (140)$$

This is the first-order weighted Stokes identity naturally associated with the Pearson pair (σ, w) . Jacobi, Gegenbauer, and Chebyshev weights require the corresponding weighted domains when w or σw is singular or vanishes at an endpoint; the algebraic identity then holds on the natural dense test space and the boundary trace is interpreted accordingly.

This distinction resolves the relation with Bochner–Pearson families. A gauge may be chosen so that its kernel line coincides with a Chebyshev, Gegenbauer, or symmetric Jacobi polynomial. Nevertheless, Theorem 8.3 prevents the gauged pairing from being the corresponding weighted moment functional. To recover the positive orthogonality, Gaussian quadrature, and Sturm–Liouville operator of the classical family, one must pass to the weighted Stokes triple (139).

14 Relation to existing theory and scope of the claims

Several neighbouring results are classical or already established.

- Gaussian nodes are the roots of orthogonal polynomials, and generalized Gaussian quadrature treats broader Chebyshev systems [24, 11, 17]. Positive moment functionals on finite-dimensional function spaces admit finite positive atomic representations by Tchakaloff's theorem [25, 2].
- SBP norm matrices carry high-order quadrature information [15]; generalized and function-space SBP make quadrature-based existence and construction explicit [10, 12], while dual-pairing SBP develops a different forward–backward operator framework [27].
- Recent FSBP work directly constructs optimal nodes through generalized Gaussian and Gauss–Lobatto quadrature [14, 3].
- Dirac structures admit gauge transformations by closed two-forms [8], and Stokes–Dirac systems possess gauge-reduction mechanisms in continuous and discrete settings [23]; the Hilbert-space domain and boundary duality required for maximality are treated analytically in [7].
- Weak-form and dual-field finite-element exterior-calculus discretizations already use power-preserving projection, duality, and representation choices [19, 6]. They provide structured constructions rather than a classification of all compatible mixed pairings for fixed (E, F, D, b) .

The present results do not claim a new derivation of Gaussian quadrature from orthogonality, nor a new general FSBP existence theorem. The principal claims are instead:

1. the compatibility and full-rank existence criteria for the abstract length-one Stokes structure (E, F, D, b) , its affine classification, and the rank-two boundary radical theorem;
2. the exact relative-open image, constructive synthesis, fibre dimension, two-component fibre topology, base topology, global continuous/smooth-section obstruction, minimal atlas, explicit two-chart synthesis, and radial blow-up theorem for the polynomial projective kernel map;
3. the hyperbolic barycentre characterization of admissible interior real degree- N root sets;
4. the moment-factorization formulation of locality and the rigidity theorem identifying $L^2/\text{Gauss–Legendre}$ as the unique local scalar representative of the polynomial gauge class;
5. the closed minimum-gauge formula, its exact global rank stratification by kernel degree, the exact attainment criterion in the full-rank fibre, the rectangular right radical, the quantitative transverse opening of the lower-degree defect, the resulting global canonical pairing section on the degree- N kernel stratum, the canonical nodal section on its simple interior real-root part, its hyperbolic stability near Gauss, and the collision obstruction to a global estimate;
6. the Riesz identification of the gauge freedom with the skew operators $K_N^* = -K_N$, $K_N|_{\ker d} = 0$, its factorization $K_N = L_N d$, and the resulting precise separation from local $P^{(0)}$ terms, boundary-port coordinate gauges, and weighted positive metrics, together with the exact coherence criterion for extension to $H^1/\mathbb{R}$.

The abstract affine theorem is elementary finite-dimensional linear algebra; generalized moment representations for Chebyshev systems are classical. The novelty claimed here lies in their joint Stokes interpretation and, more specifically, in the exact polynomial kernel image and fibre theorems, locality rigidity, and the associated gauge/operator distinction. To the best of the author's knowledge, this combination has not appeared in the SBP, generalized quadrature, or Stokes–Dirac literature.

15 Conclusion

For a surjective length-one discrete Stokes structure, compatibility is equivalent to the vanishing of the boundary form on $\ker D \times \ker D$. The compatible pairings then form an affine space directed by $\Lambda^2(E/\ker D)^*$; full rank is generic whenever it is allowed by the parity–boundary criterion. Stokes therefore fixes the symmetric boundary power but leaves a genuine two-form freedom on the derivative quotient.

For $\mathbb{P}_N \xrightarrow{d/dx} \mathbb{P}_{N-1}$, the radical of a full-rank pairing is an intrinsic, gauge-dependent polynomial line. The kernel map has an exact relative semialgebraic image, constructive fibres of dimension $(N-1)(N-2)/2$, and two Pfaffian components for $N \geq 3$. On the simple interior real-root stratum, admissibility becomes

$$\sum_i \operatorname{artanh}(x_i) = 0,$$

so admissible node sets form a codimension-one family whose equal-weight hyperbolic barycentre is the origin.

These intrinsic roots are not automatically quadrature nodes. They provide exact nodal coordinates with a full coupling matrix. Scalar locality is a separate moment condition; because differentiated products span $\mathbb{P}_{2N-1}$, it collapses the affine class to the unweighted L^2 pairing and uniquely selects the Gauss–Legendre nodes and positive Gauss weights. Thus Gauss–Legendre is not imposed by Stokes alone, but is rigid within the fixed unweighted Stokes geometry once local scalar power evaluation is required.

After the L^2 quotient metric is chosen, the minimum-norm solution of a prescribed-kernel constraint is represented by the explicit rank-two operator (83). The associated mixed pairing has rank exactly N on the degree- N stratum and $N-2$ on every nonempty lower-degree stratum, where its full left and right radicals are explicit. The defect opens transversely under a Legendre-direction perturbation. On the degree- N stratum this gives a global canonical pairing section, whose nodal restriction is locally Lipschitz near Gauss but cannot satisfy a uniform global estimate without a node-separation or conditioning factor.

The normalized kernel image is $\mathbb{R}^{N-1}$ in odd degree and $\mathbb{R}^{N-1} \setminus \{0\}$ in even degree. Although every admissible kernel has an individual full-rank realization, a global smooth realization over the entire image exists only for odd N and for $N = 2, 4, 8$. In even degree this is precisely the almost-complex obstruction on S^{N-2} . The case $N = 2$ is trivial, while the exceptional sections for $N = 4$ and $N = 8$ come from the three- and seven-dimensional vector cross products. At all other even orders, one global chart is impossible but the explicit Rodrigues construction gives an optimal two-chart atlas. This is a topological obstruction to choosing the gauge, not an obstruction to the existence of the fibres. Independently of that topology, every even-order synthesis is analytically unbounded near the deleted constant class: the radial equation forces $\|r_z\| \geq 2/\|z\|$ and hence $\|M_z\|_F \gtrsim \|z\|^{-1}$. The obstruction is therefore present even in the exceptional orders $N = 2, 4, 8$.

Finally, the discrete gauge is a quotient-compatible, generally nonlocal skew operator that factors through differentiation. Coherent uniformly bounded families extend to the continuous $H^1/\mathbb{R}$ model. This separates the pairing gauge from boundary-port coordinate changes, local $P^{(0)}$ terms, and weighted Hamiltonian geometries. Classifying which coherent nonlocal gauges survive the continuous limit is the natural next question.

A Basis covariance

Let T_E and T_F be nonsingular changes of basis in E_N and F_N . The mixed and derivative matrices transform as

$$M \mapsto T_E^\top M T_F, \quad D \mapsto T_F^{-1} D T_E.$$

Therefore

$$MD \mapsto T_E^\top(MD)T_E,$$

and the gauge two-form transforms by congruence. The coefficient vector of κ_β changes contragrediently, but the polynomial and its zero set do not. The affine classification, endpoint condition, and kernel map are basis-independent.

B Constructive synthesis for a prescribed kernel

Given a nonconstant $\kappa \in \mathbb{P}_N$ satisfying $\kappa(1) = (-1)^N \kappa(-1) \neq 0$, the proof of Theorem 6.1 gives an explicit construction. For $N = 1$, the only projective kernel is $[x]$ and the unique compatible pairing is the zero-gauge L^2 representative. Assume below that $N \geq 2$.

1. Decompose $\kappa = \alpha + \beta x + z$ with $z \in (1 - x^2)\mathbb{P}_{N-2}$.
2. If N is even, choose a skew matrix C on Z_N with $\ker C = \text{span}\{z\}$ and choose r satisfying $z^\top r = 2\alpha$.
3. If N is odd, choose any invertible skew matrix C and set $r = \beta^{-1}Cz$.
4. Form $A_{r,C}$ from (32). Because $A_{r,C}(\cdot, 1) = 0$, the form factors through the derivative in its second slot; surjectivity of $d : E_N \rightarrow F_N$ then gives a unique mixed matrix M such that $MD = A_{r,C}$.

The resulting M is full rank, Stokes-compatible, and has left radical $\text{span}\{\kappa\}$. This construction turns the kernel image theorem into an exact synthesis procedure rather than a dimension count.

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